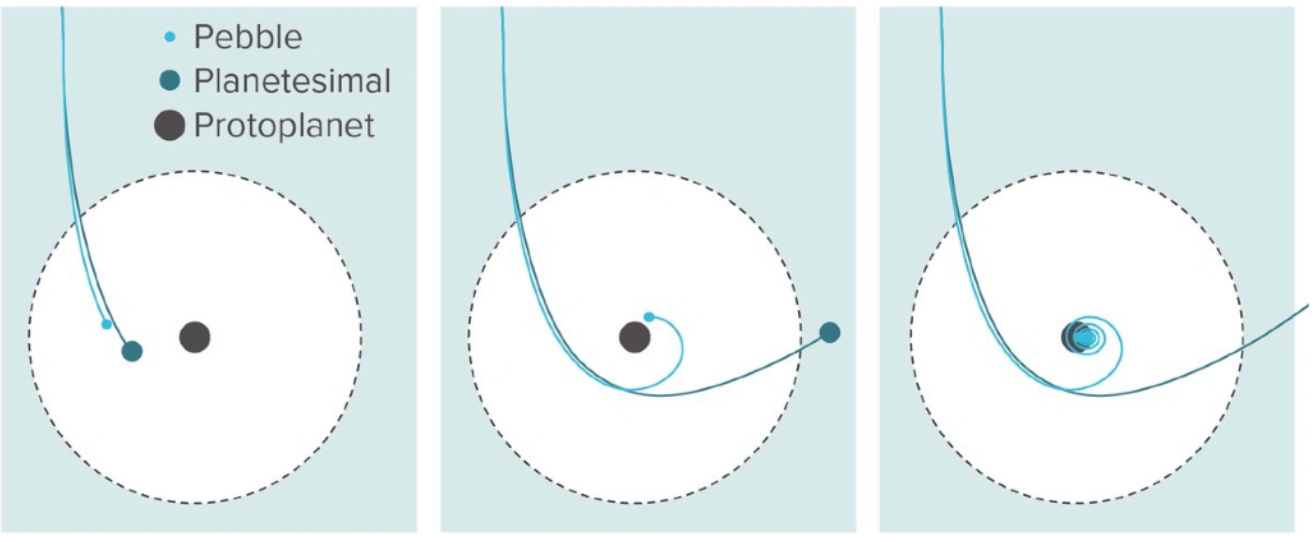
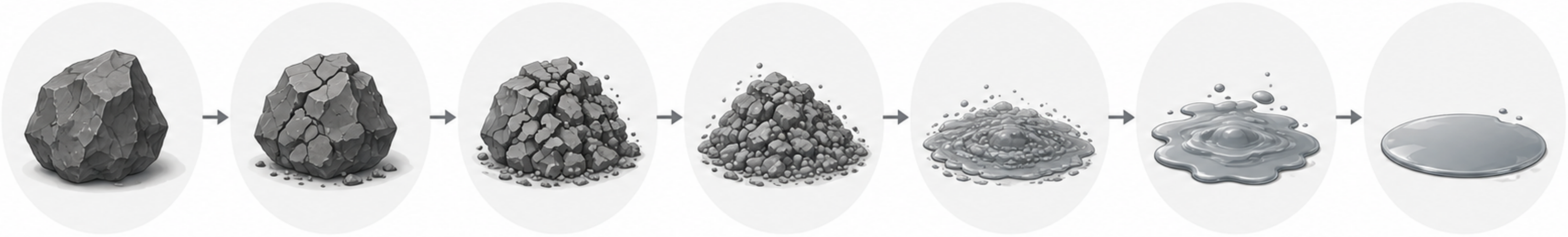
A multi-fluid approach to pebble accretion





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TU Delft | Planetary Exploration

erc DUSTSPEC

Multi-fluid pebble accretion

Implications of this talk

1. Pebble accretion is possible with a fluid approach
2. Pebble distribution changes gas dynamics drastically
3. After isolation, solid-to-gas ratios can reach unity values, which could cause instabilities

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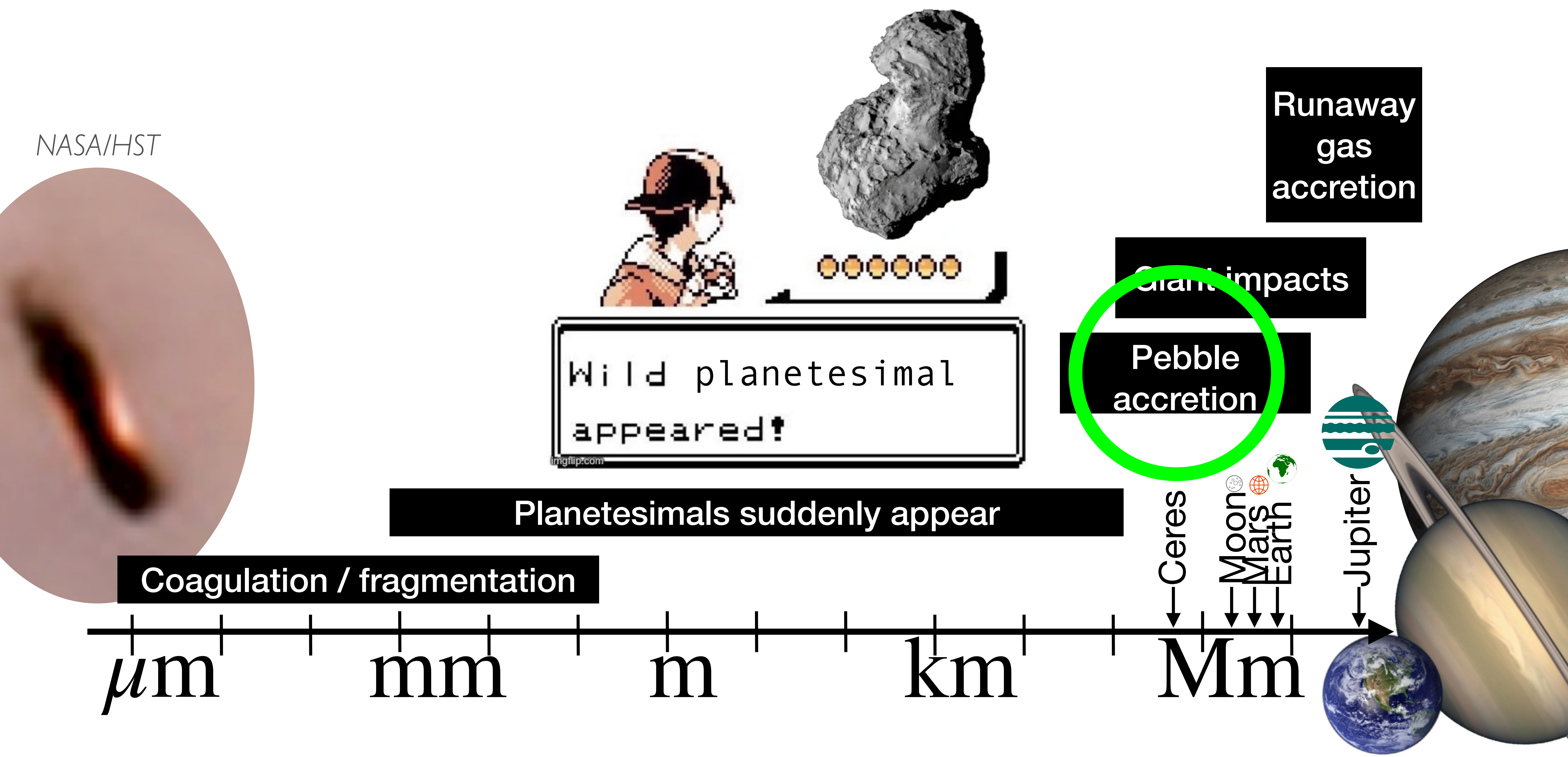
Required slide

From μm to gas giant



First things first

A simple disc

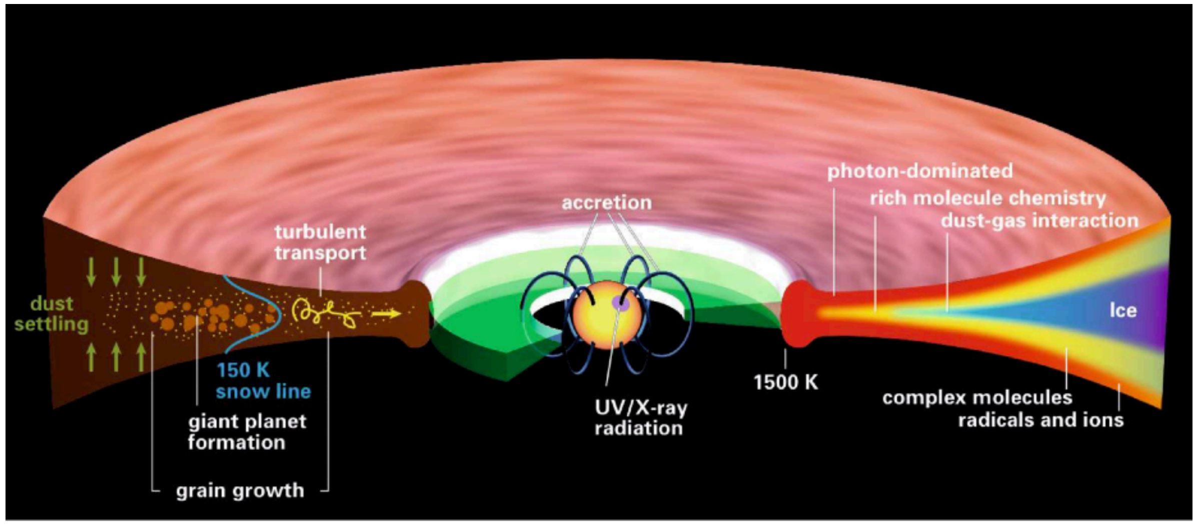
- Gas moves sub-Keplerian

$$v_{\mathrm{gas},\varphi} = (1-\eta)\; v_{\mathrm{K}}$$

$$\eta = -\frac{1}{2}\frac{c_{\mathrm{s}}^2}{v_{\mathrm{K}}^2}\frac{\mathrm{d}\ln P}{\mathrm{d}\ln \mathrm{r}} \approx \left(\frac{H}{r}\right)^2$$

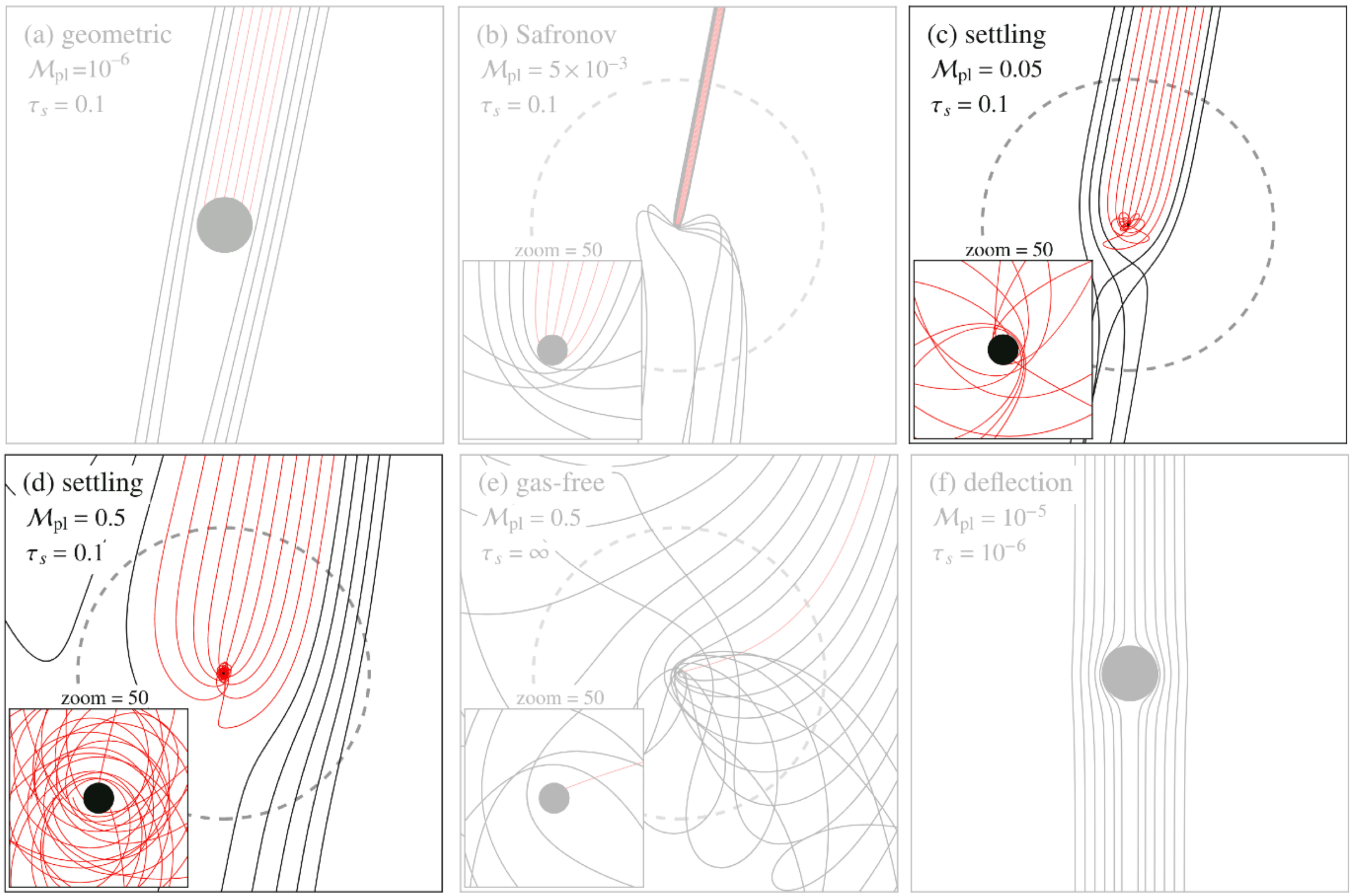
- Steady drift solution for solids

$$v_{\mathrm{dust},r} = -\frac{2\mathrm{St}}{\mathrm{St}^2+1}\eta v_{\mathrm{K}}$$



[Henning & Semenov 2013]

Pebble accretion



[Ormel 2017]

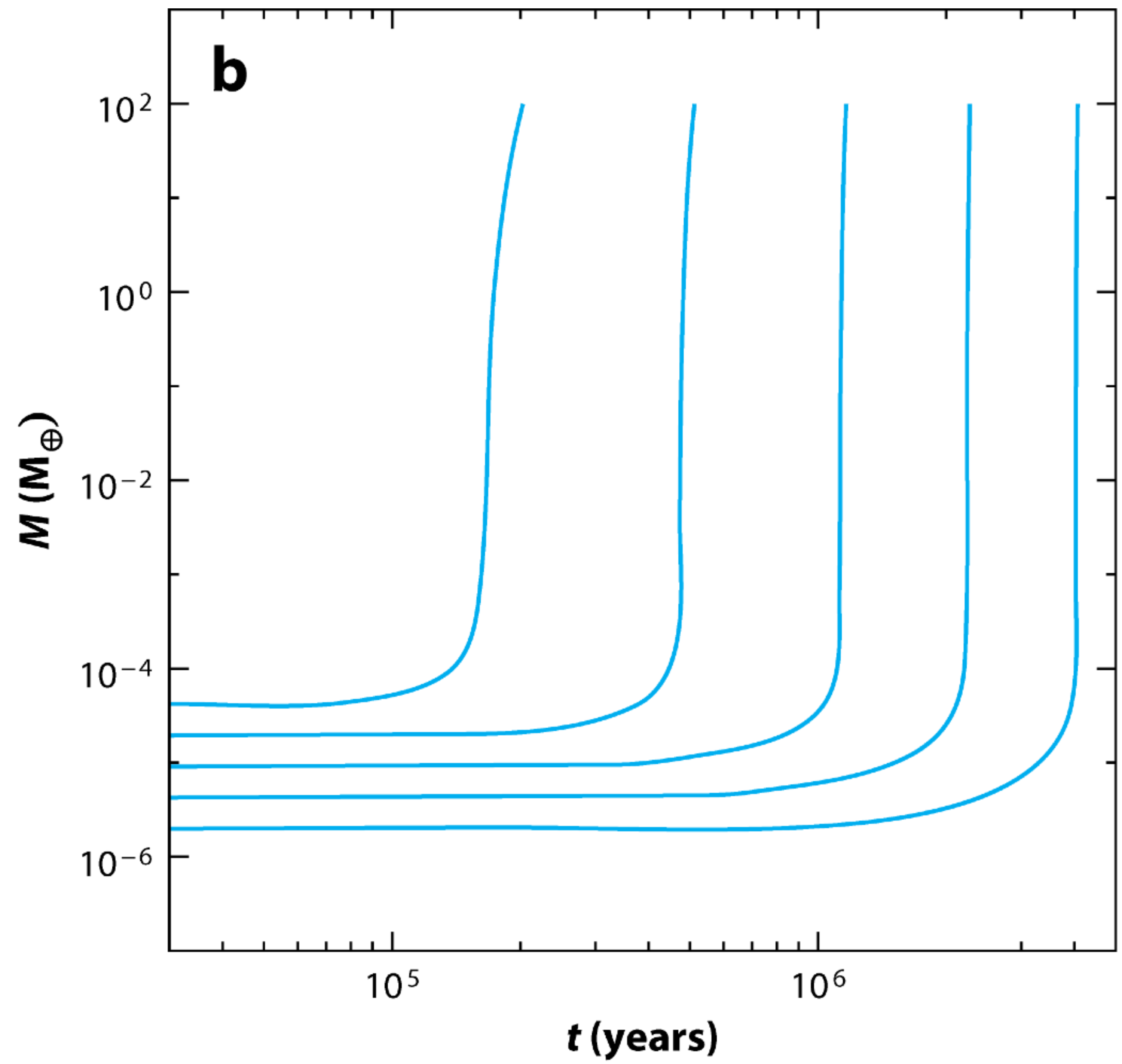
3 requirements for PA:

- Enough mass:
  $M_{\rm pl} \tilde{\in} \left[10^{-3}, 10^{1}\right] M_{\oplus}$
- **Gas needs to be present**
  $t_{\rm disk} \tilde{<} 10\ \rm Myr$
- **Solids need to be "pebble" size**
  $\rm St \tilde{\in} \left[10^{-3}, 10^{0}\right]$

Pebble accretion

A planet(ary core) can be created within lifetime of disc:

- Spiraling nature of pebbles -> large accretion area
- Incoming mass supply of drifting pebbles



[Johansen & Lambrechts 2017]

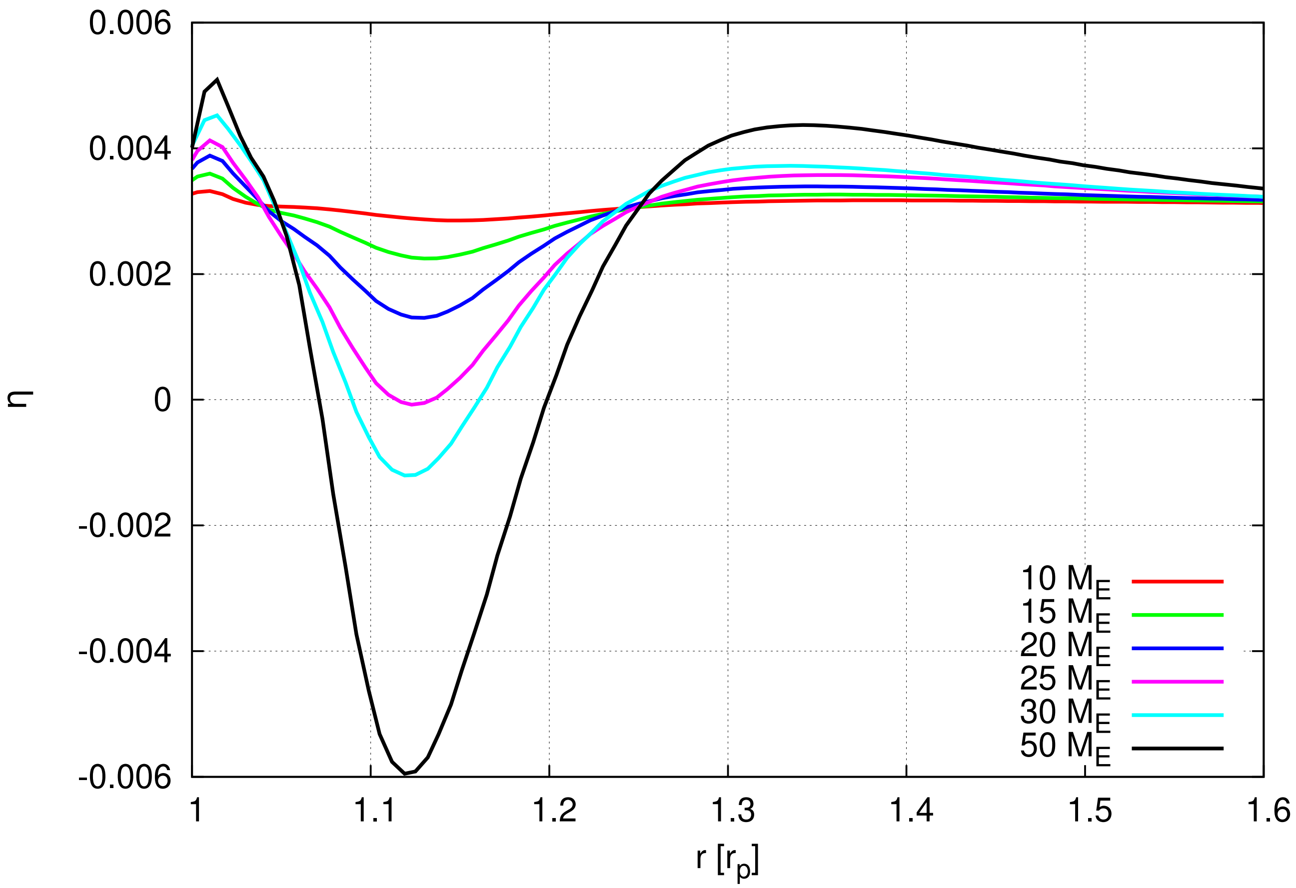
Pebble accretion

A planet(ary core) can be created within lifetime of disc:

- Spiraling nature of pebbles -> large accretion area
- Incoming mass supply of drifting pebbles

Too massive?
Incoming mass halted:

$$v_{\mathrm{gas},\varphi} = (1 - \eta)\ v_{\mathrm{K}}$$



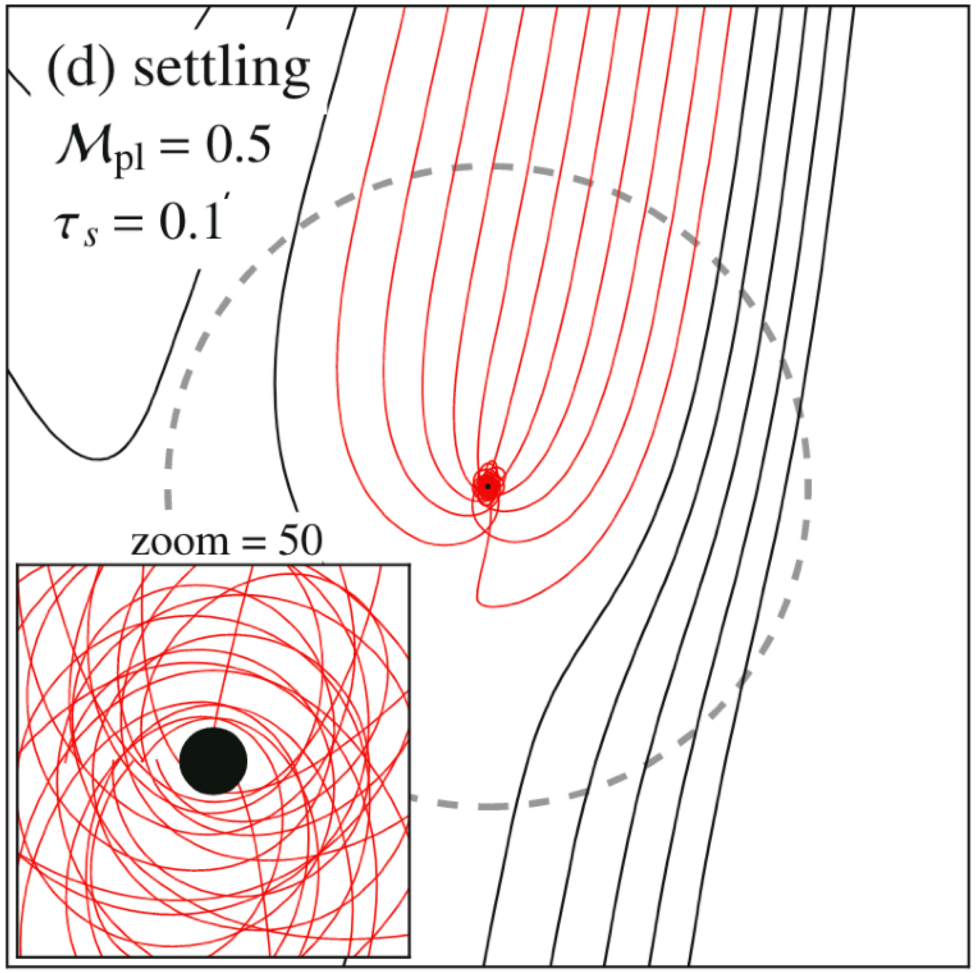
[Bitsch et al. 2018]

Pebble accretion

Via particle method

Looks great!

- Planets are made fast
- Assumes:
    - Unperturbed gas disc (static)
    - Single pebble size (monodisperse)
- What if planet becomes large enough to perturb disc?



[Ormel 2017]

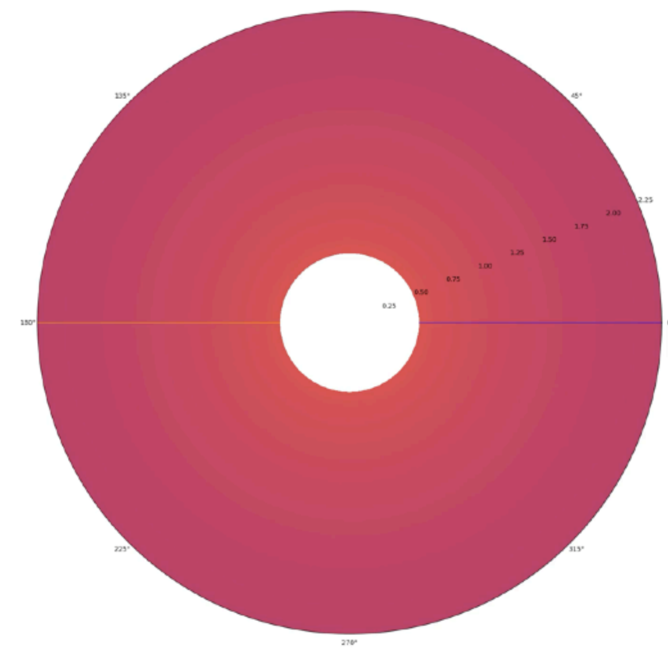
Pebble accretion via fluid method

A few advantages:

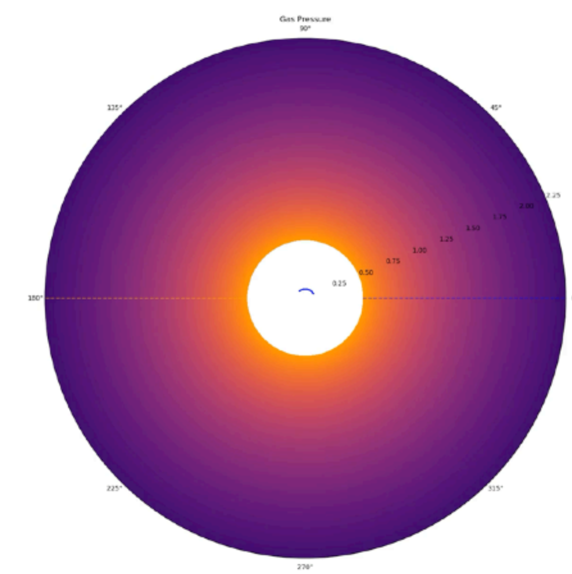
- Multiple dust species are possible (polydisperse)
- Dust feedback can be turned on
- Fluid method can incorporate planets perturbation (isolation can be seen)
- But can PA be done as a fluid?

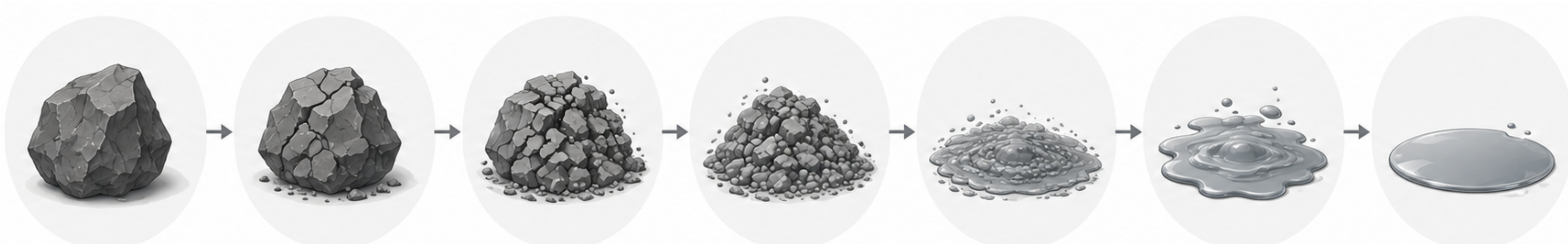
(movie)

Dust



Gas





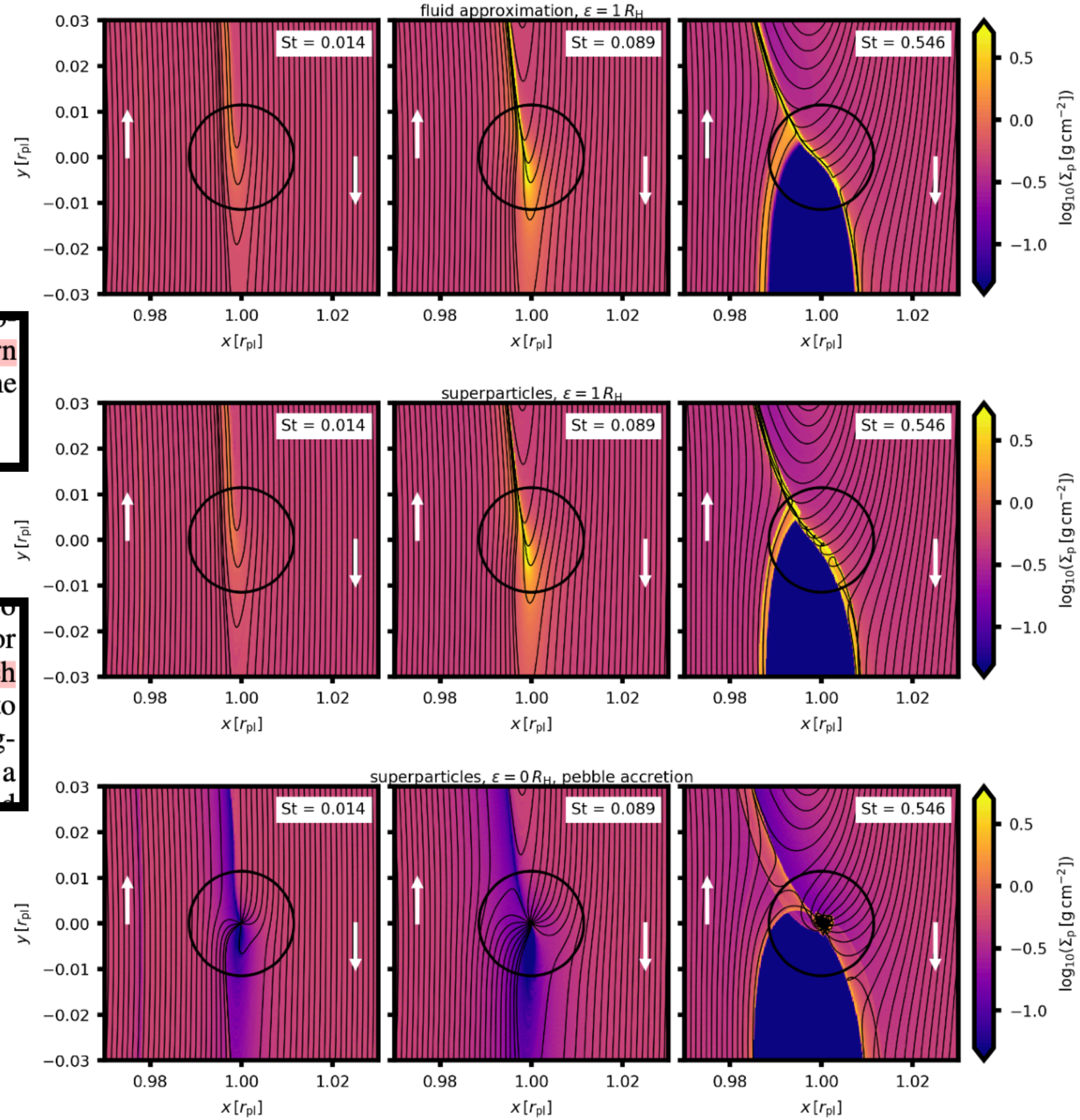
Pebble accretion via fluid method

They said it couldn't be done

> (any spurious artefact in the front-rear asymmetry of the peb-
> ble distribution would change the torque) and therefore, we warn authors against combining the material sink approach with the 2D potential smoothing for pebbles.

> the planet. However, due to the 2D limitation, these studies also employ the smoothing of the planetary gravitational potential for the pebble fluid. In the light of our results, such an approach is dangerous. Consider the first row in Fig. 3 corresponding to multi-fluid simulations with the potential smoothing and imagine that the pebble density is reduced at each time step within a sink area similar to the Hill sphere. This reduced density would

We tried it anyway

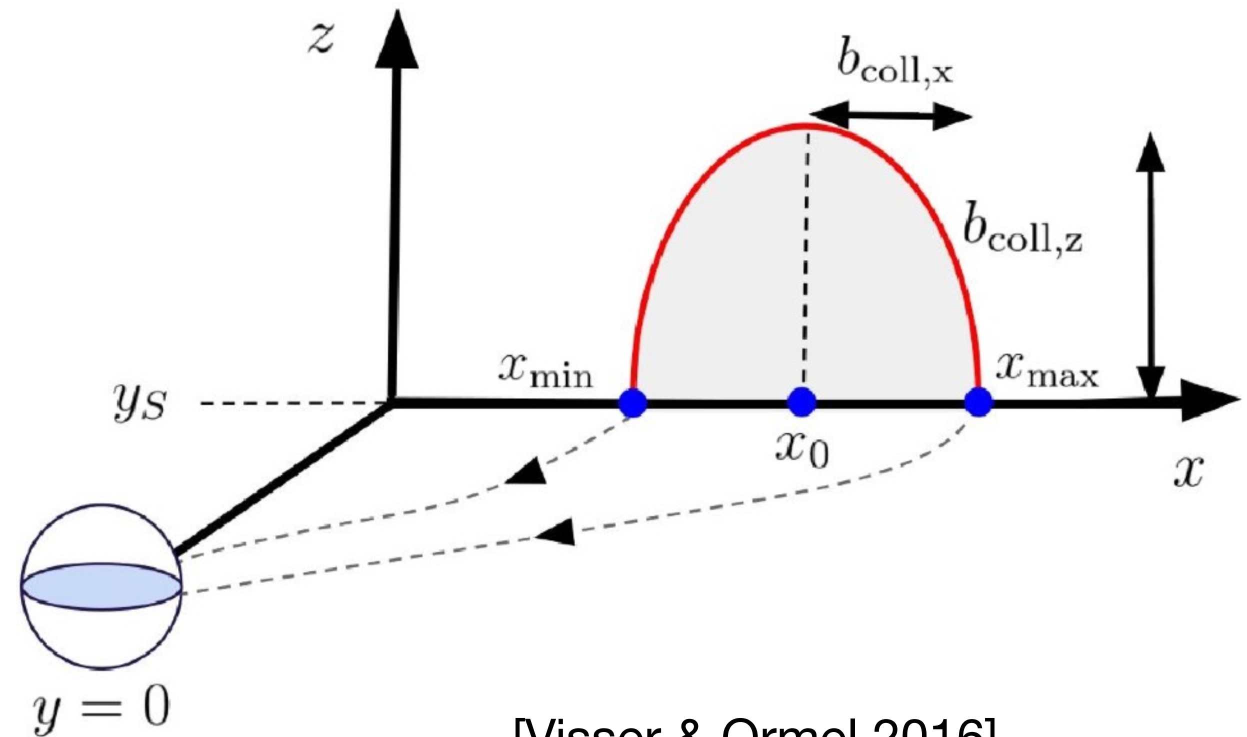


[Chrenko et al. 2024]

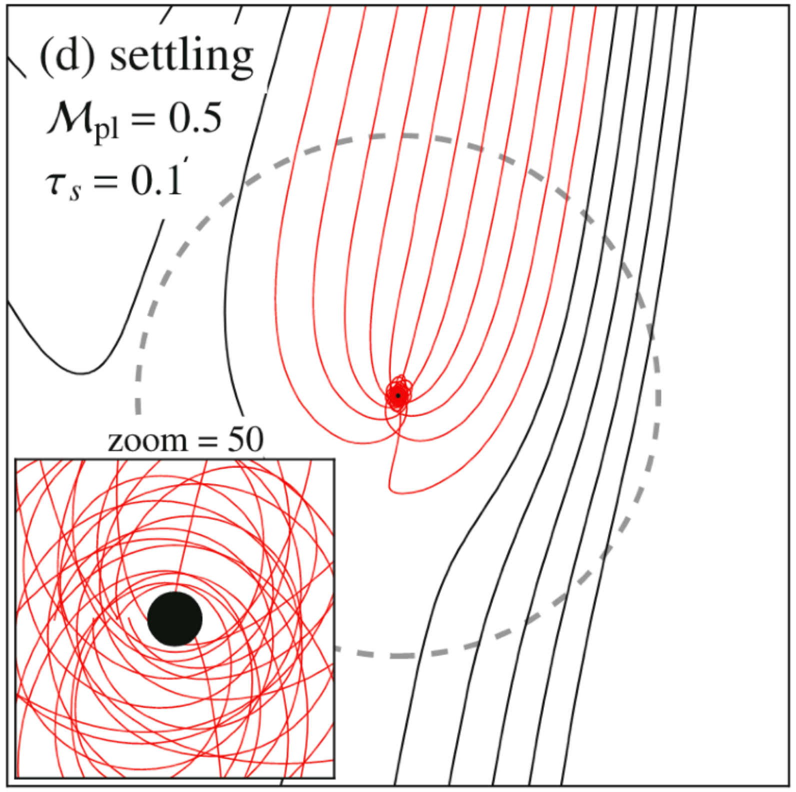
Pebble accretion via fluid method

- All mass within b_{coll} is accreted
- We use FARGO3D, and “delete” all mass in grid cells within b_{coll}
- Smoothing factor λ:

$$\lambda_{\mathrm{gas}} = 0.6H_{\mathrm{gas}} \qquad \lambda_{\mathrm{dust}} = 0$$



[Visser & Ormel 2016]



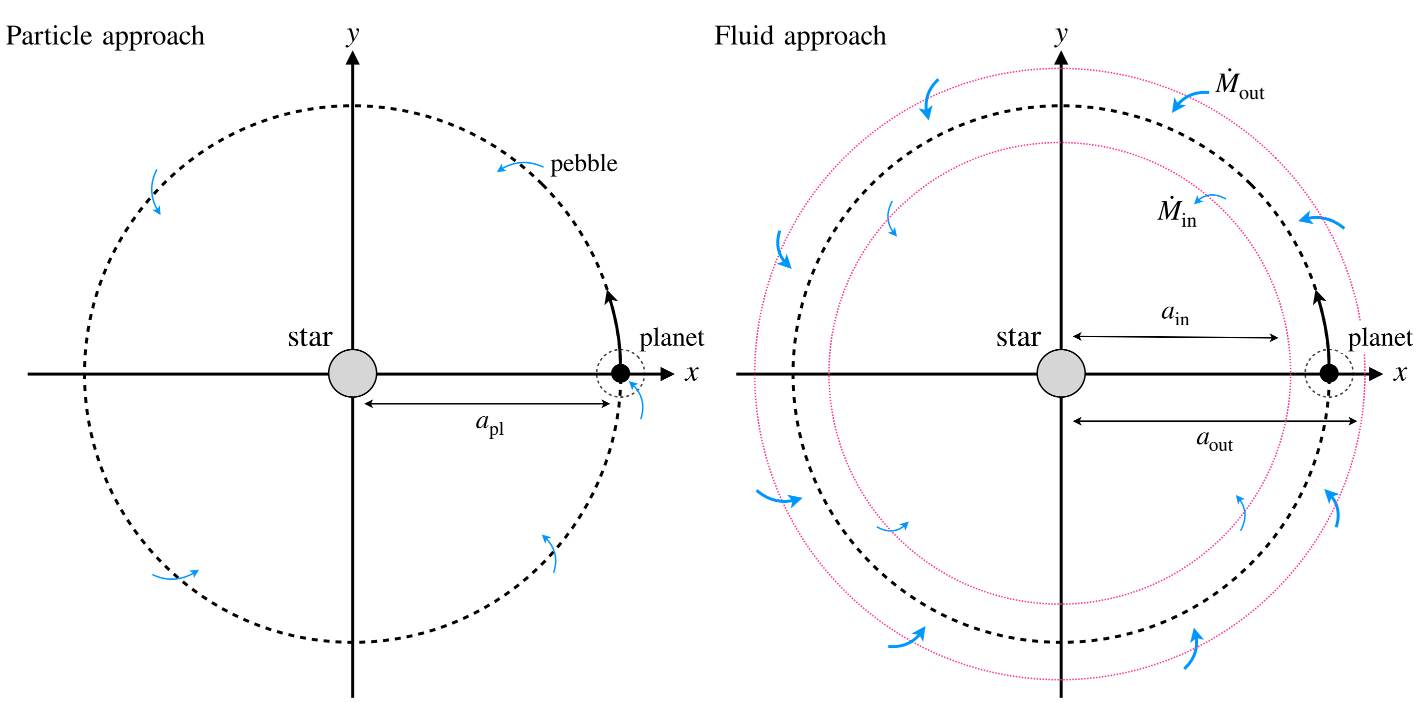
[Ormel 2017]

Pebble accretion via fluid method

Let's test it

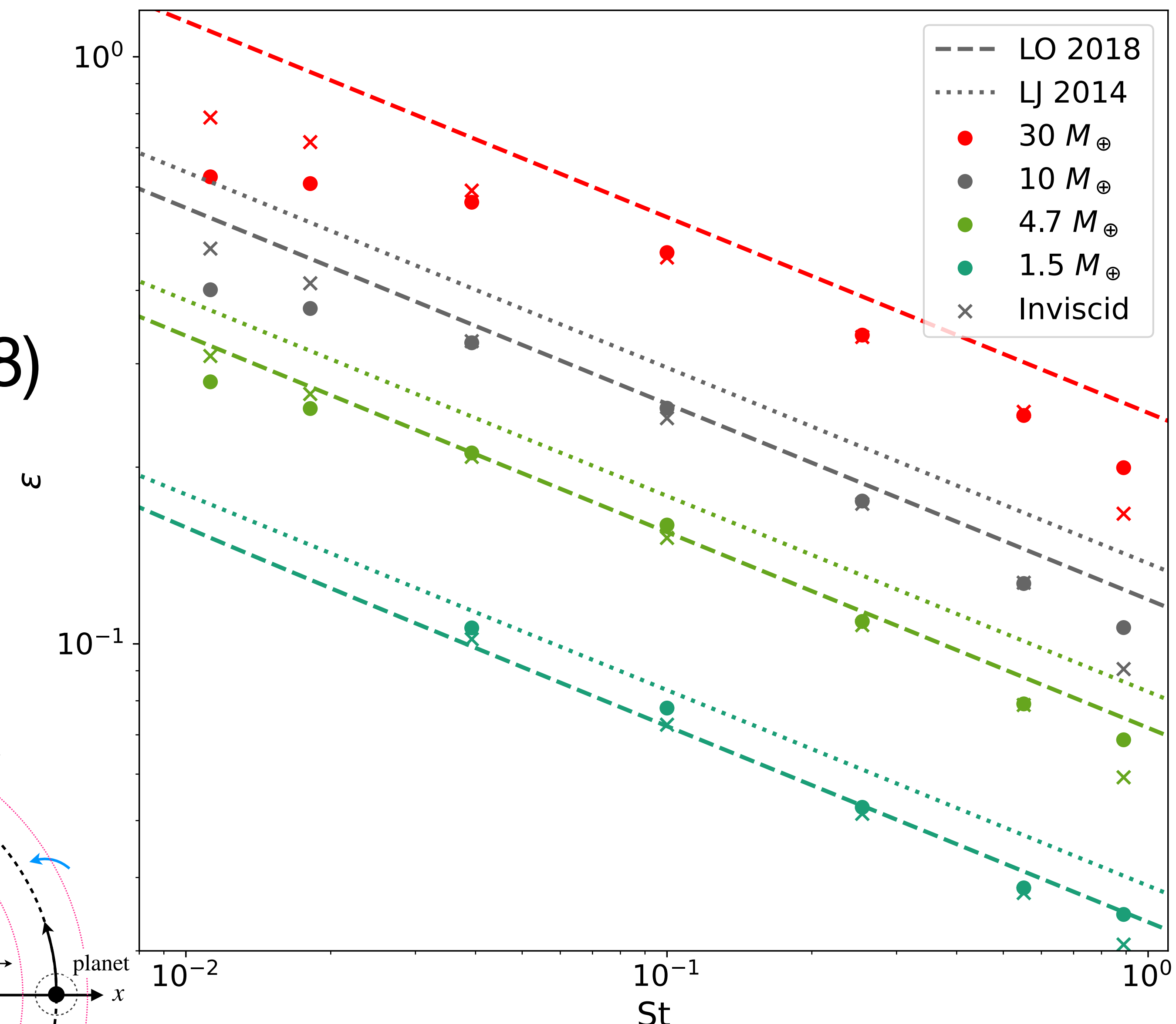
- Comparing our method with previous work (Liu & Ormel 2018)

  Via accretion efficiency ε



$$\varepsilon = \frac{N_{\rm hit}}{N_{\rm total}}$$

$$\varepsilon = \frac{\dot{M}_{\rm out} - \dot{M}_{\rm in}}{\dot{M}_{\rm out}}$$



[Konijn & Paardekooper 2026]

Pebble accretion via fluid method

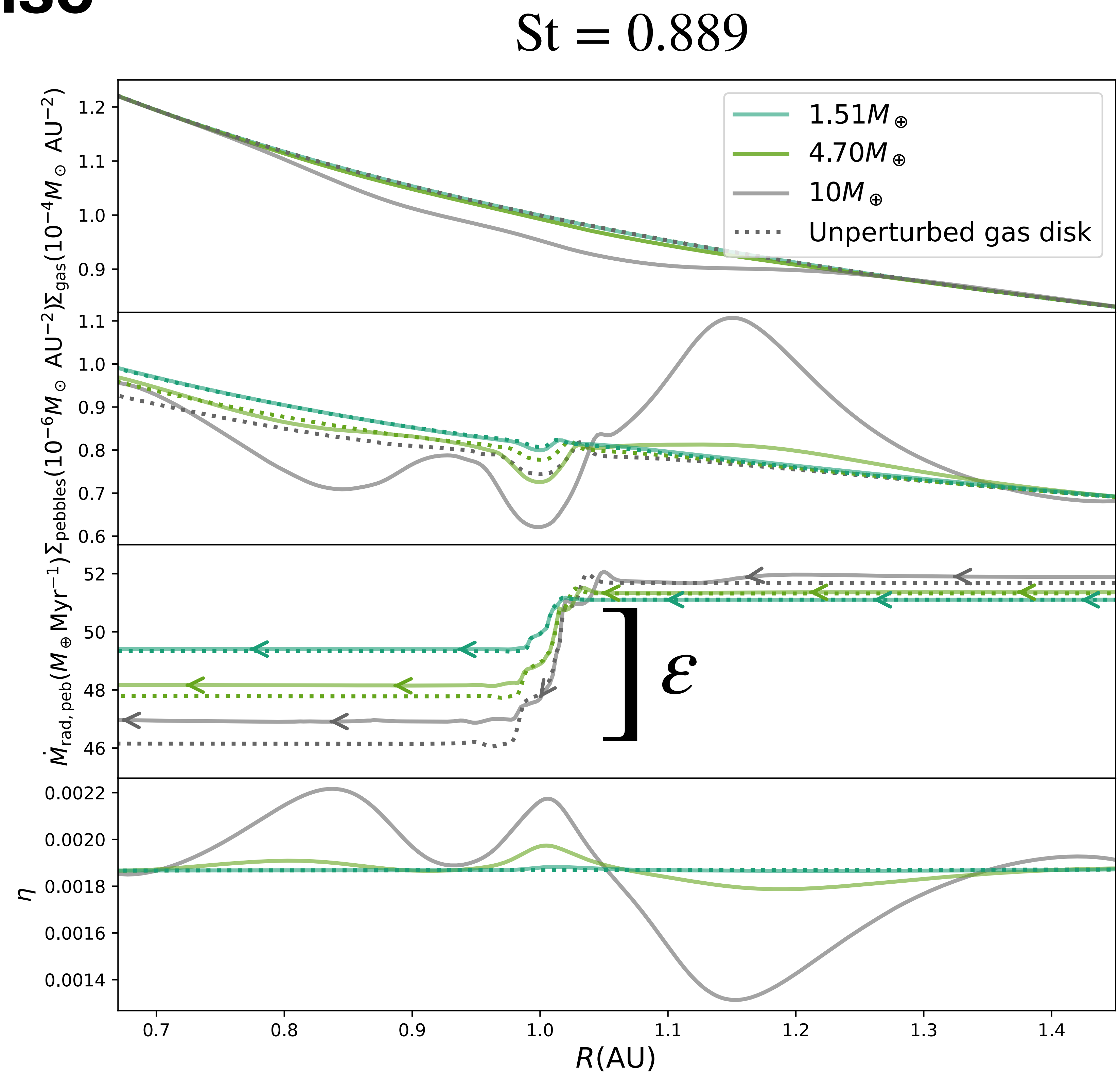
What can we do?

1. Perturbing the disc
2. Insert a polydisperse pebble population
3. Watch the isolation process

Pebble accretion via fluid method

1. Perturbing the disc

- First effect $\gtrsim 5M_\oplus$
- Efficiency is lower when gas disc is perturbed by planet (for $\mathrm{St} \gtrsim 0.3$)
- Effect increases with planet mass



Pebble accretion via fluid method

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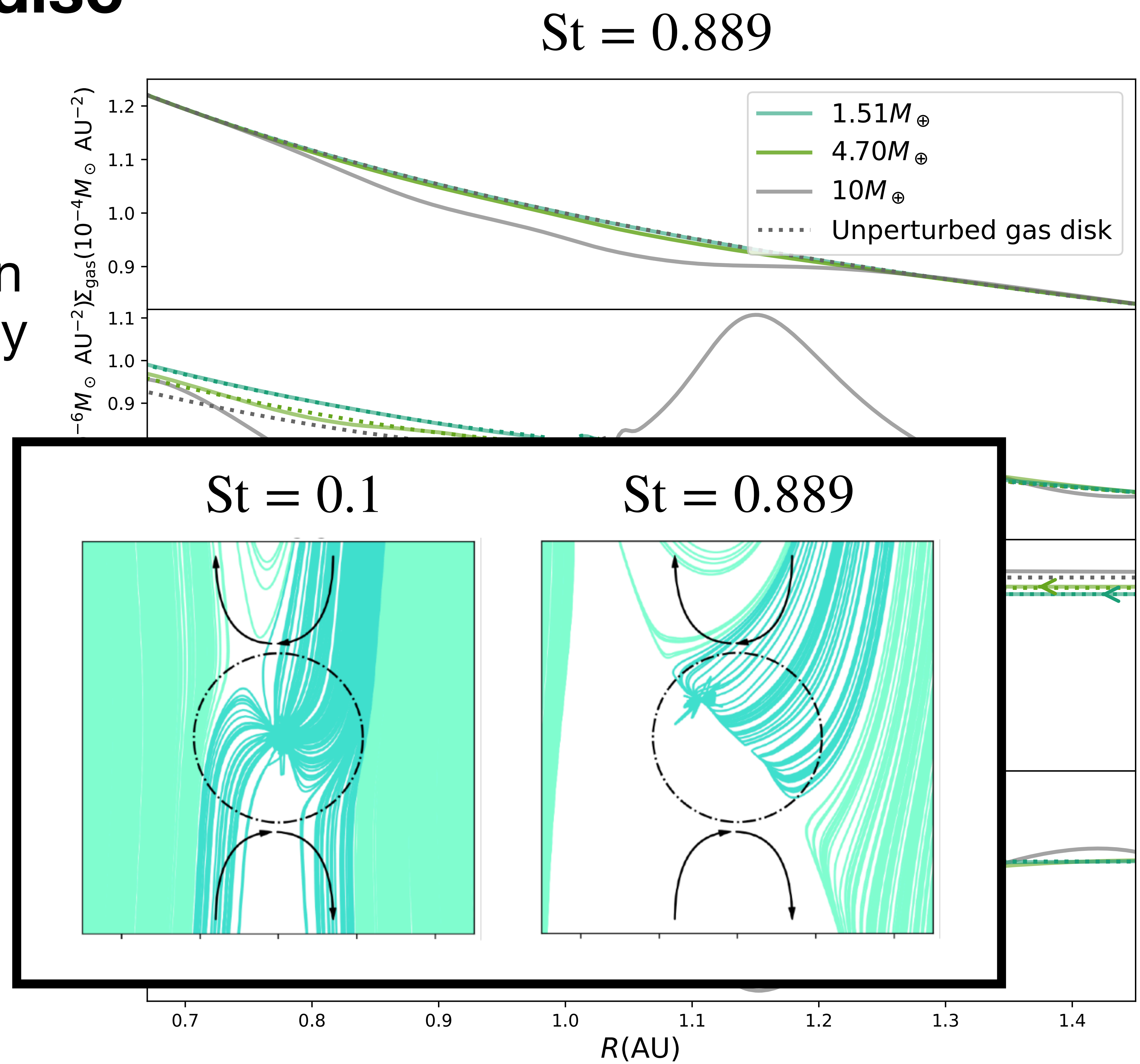
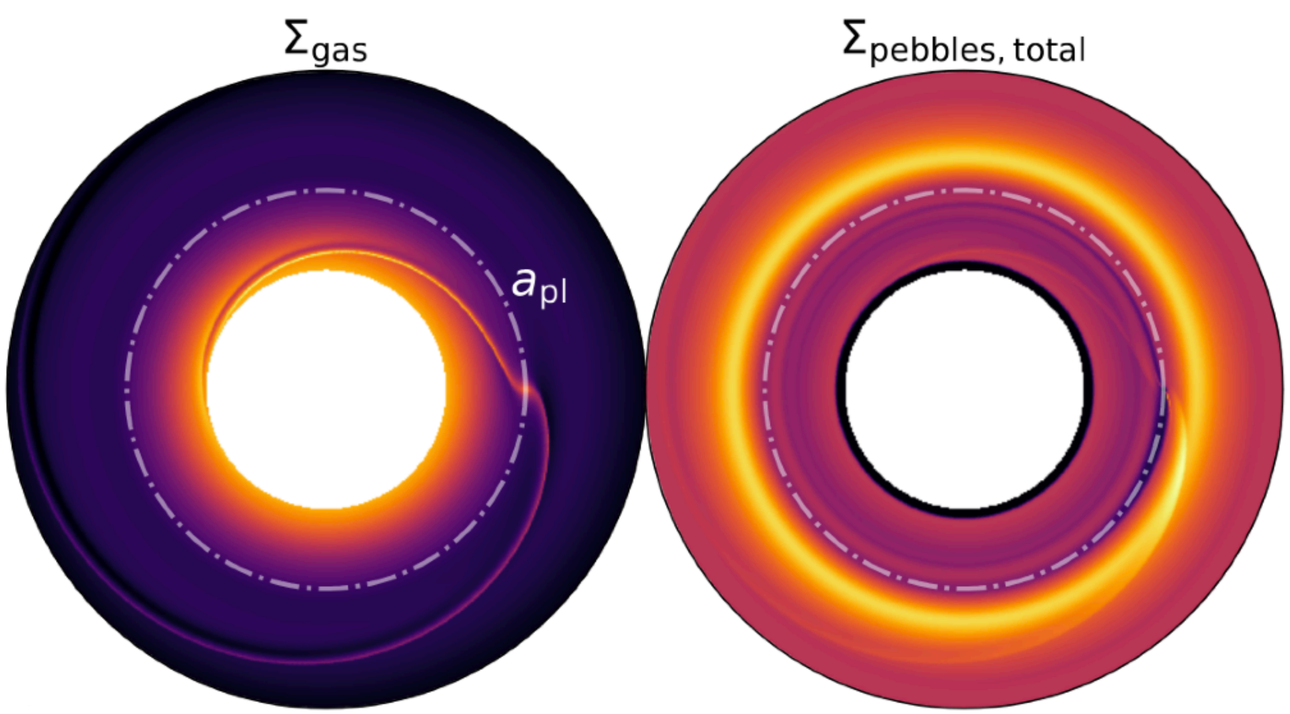


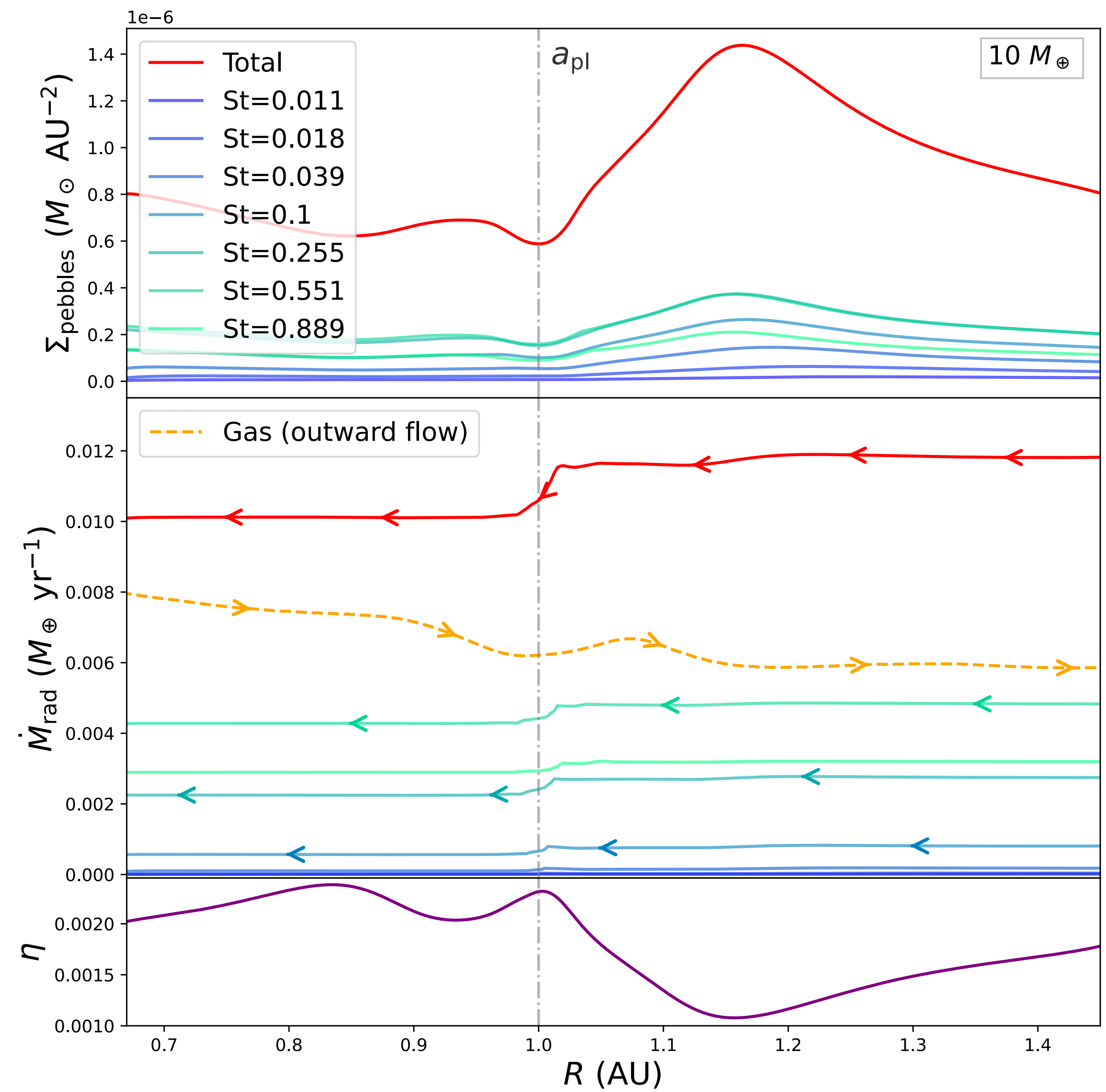
Pebble accretion via fluid method

2. Insert a polydisperse pebble population

Fiducial (polydisperse) model:

- Polydisperse, perturbed, viscous disk, with $10M_\oplus$ planet





[Konijn & Paardekooper 2026]

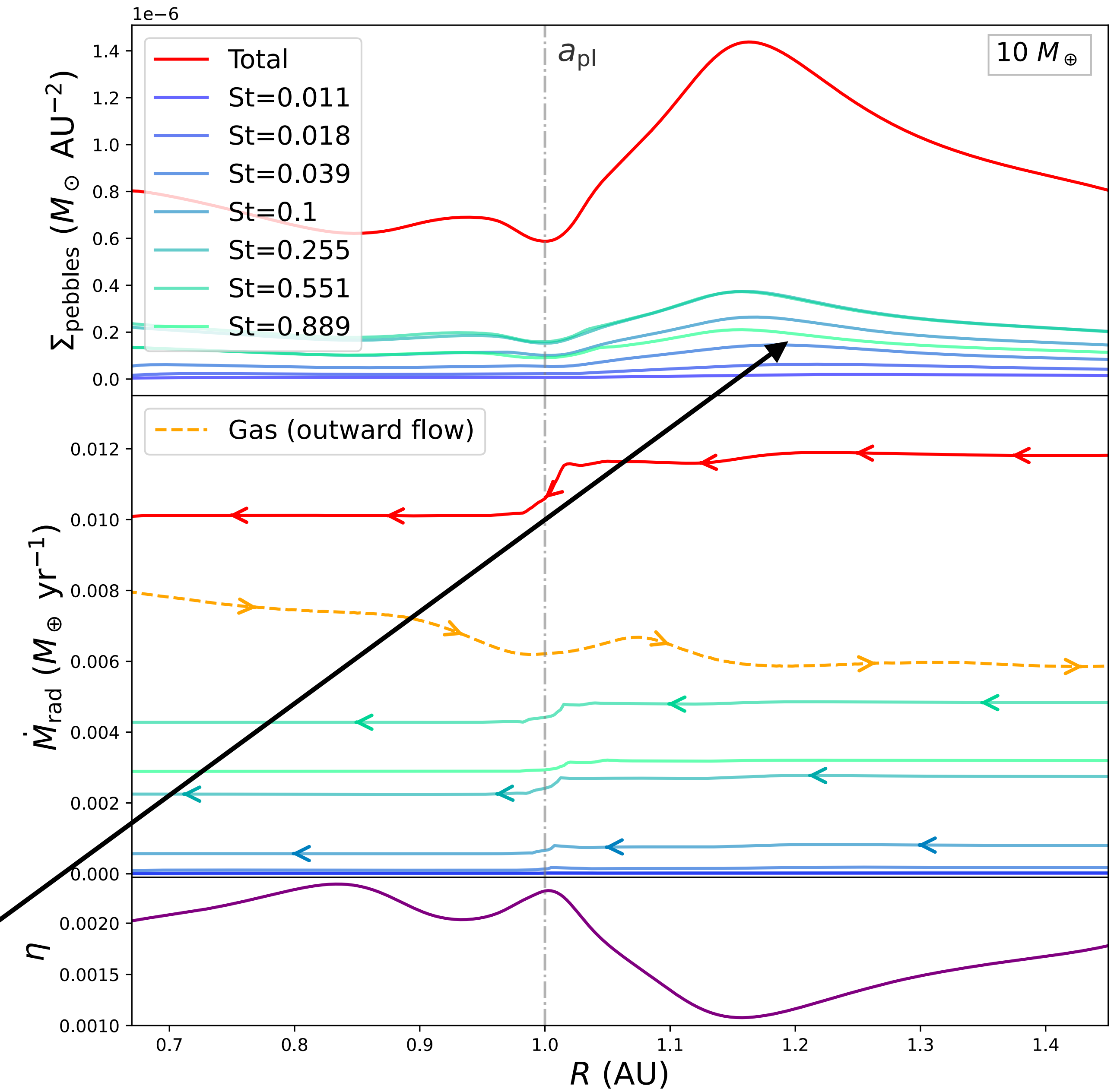
Pebble accretion via fluid method

2. Insert a polydisperse pebble population

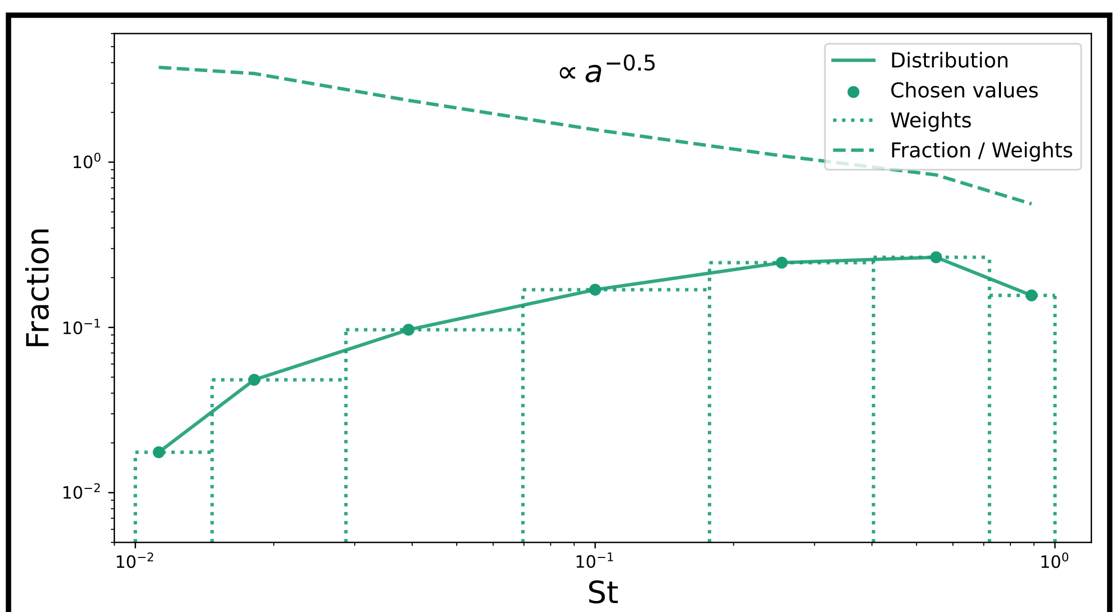
Fiducial (polydisperse) model:

- Discrete distribution acts like a *continuous* MRN-distribution via GL-conjecture
- Computationally:
  Drag matrix instead of drag integral

$$\frac{1}{\Sigma_\mathrm{g}}\int \sigma \frac{\vec{v}_\mathrm{d}-\vec{v}_\mathrm{g}}{\mathrm{St}}\mathrm{dSt} \approx \frac{1}{\Sigma_\mathrm{g}}\sum_n w_n \sigma(\mathrm{St}_n)\frac{\vec{v}_\mathrm{d}(\mathrm{St}_n)-\vec{v}_\mathrm{g}}{\mathrm{St}_n}$$



[Konijn & Paardekooper 2026]



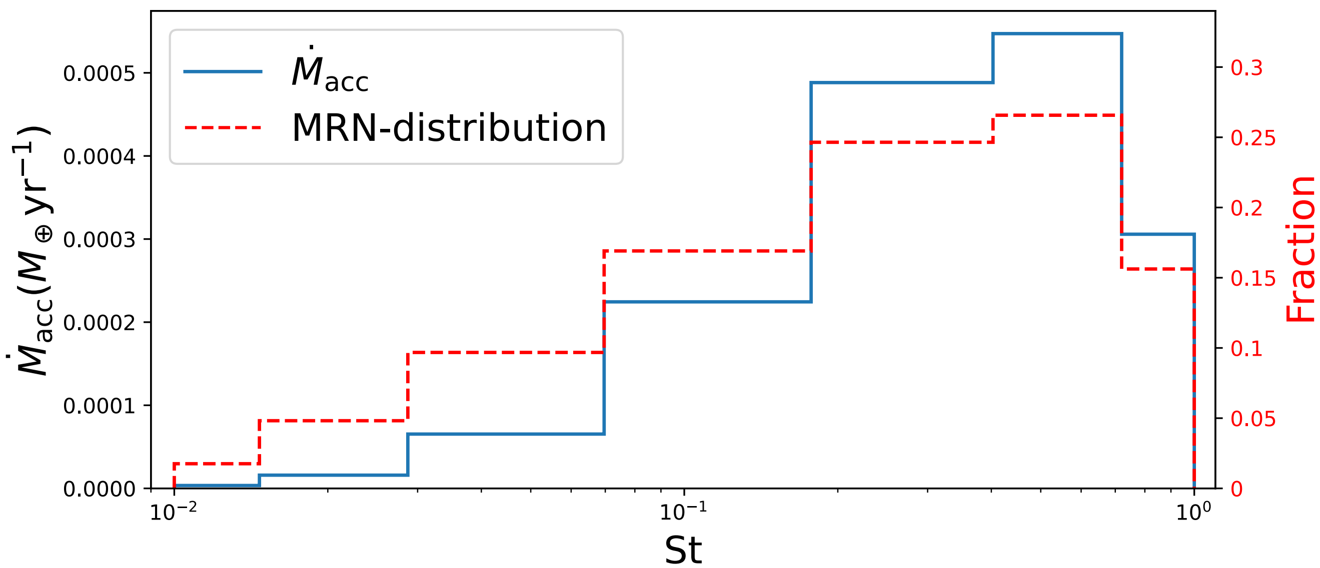
The density distribution of the different Fluids

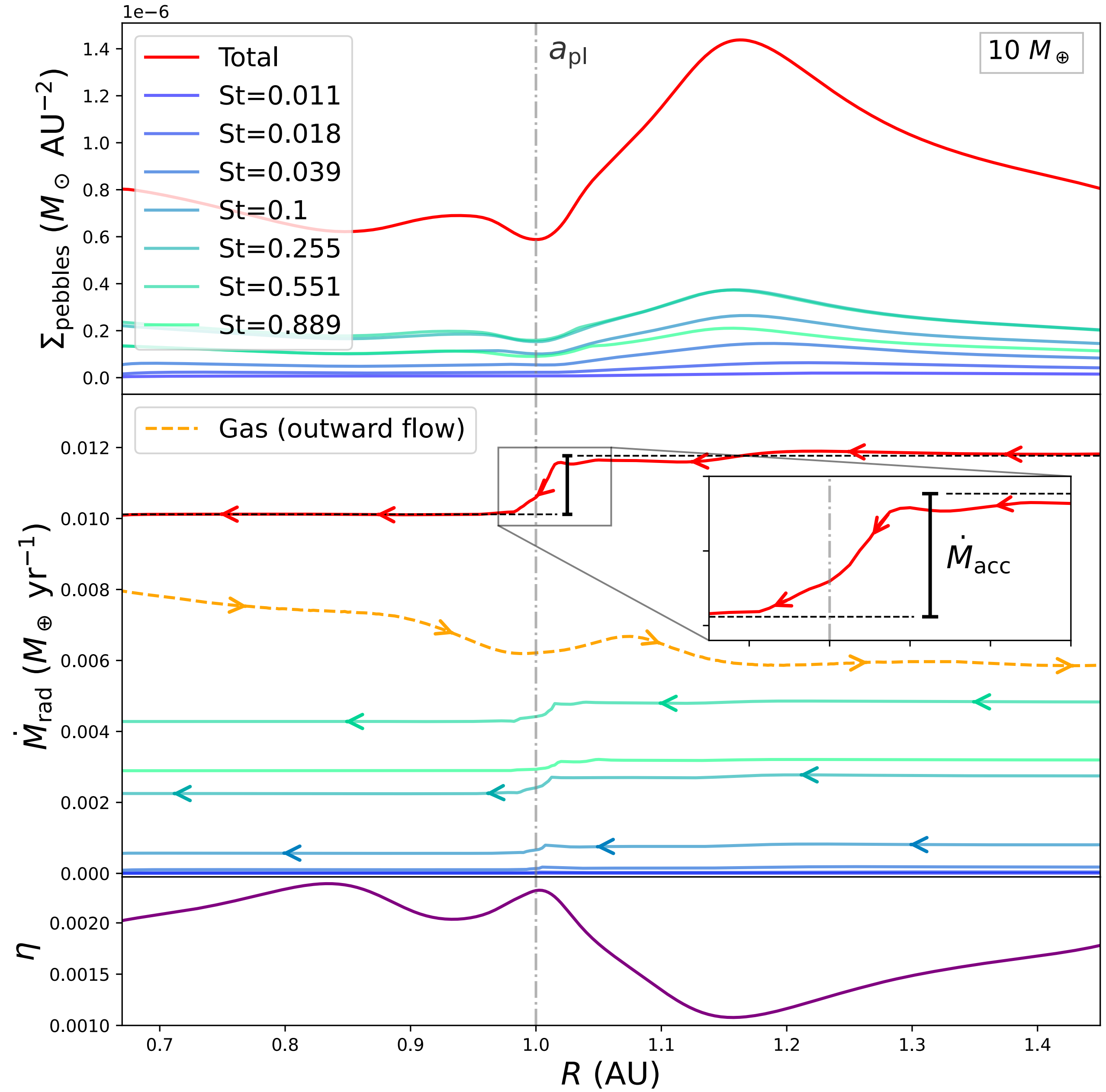
Pebble accretion via fluid method

2. Insert a polydisperse pebble population

Fiducial (polydisperse) model:

- Accretion per pebble species





[Konijn & Paardekooper 2026]

Pebble accretion via fluid method

2. Insert a polydisperse pebble population

- How does a polydisperse population compare to analytical results?

An Analytical Theory for the Growth from Planetesimals to Planets by Polydisperse Pebble Accretion

Wladimir Lyra[1], Anders Johansen[2,3], Manuel H. Cañas[1], and Chao-Chin Yang (楊朝欽)[4]

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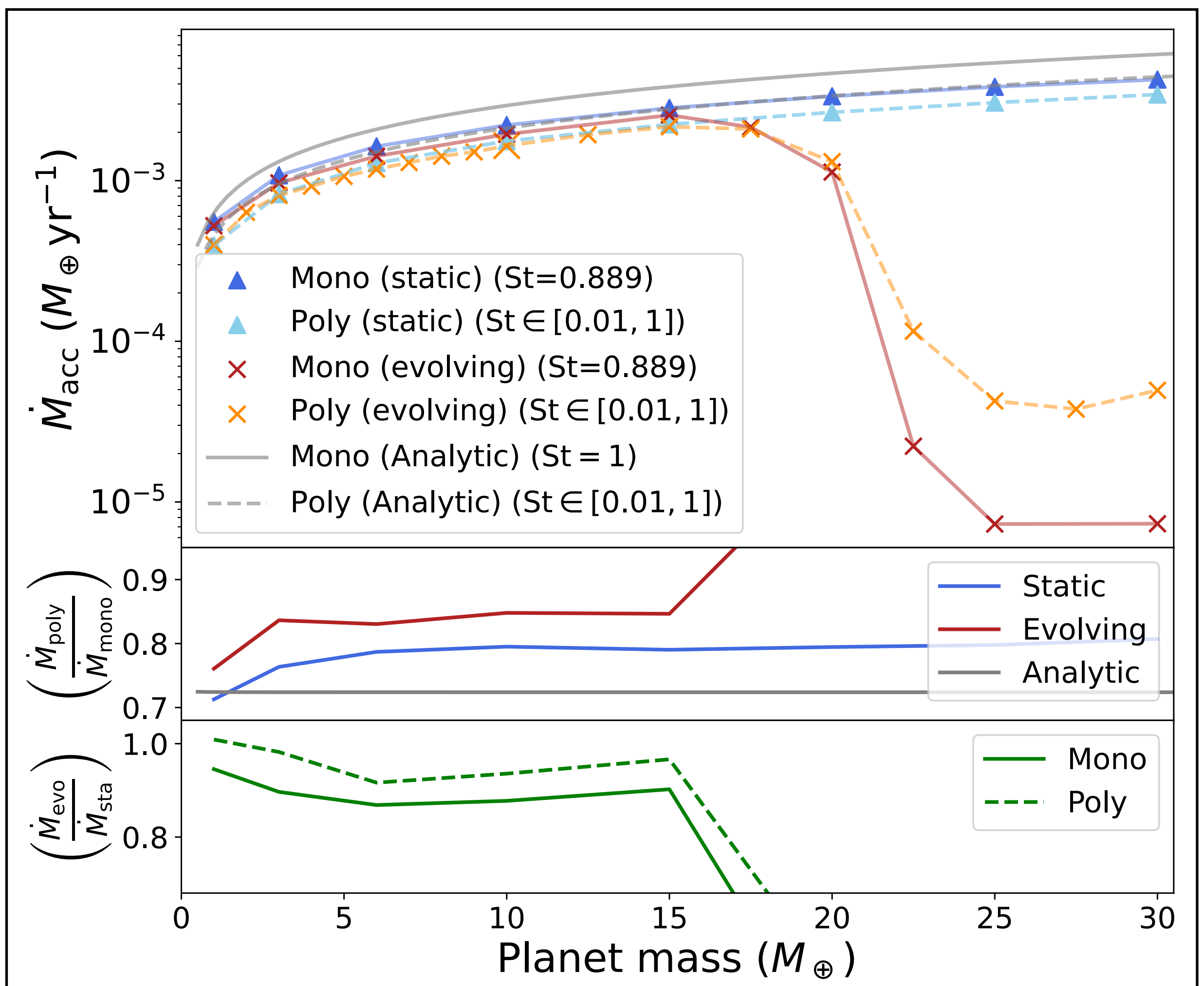
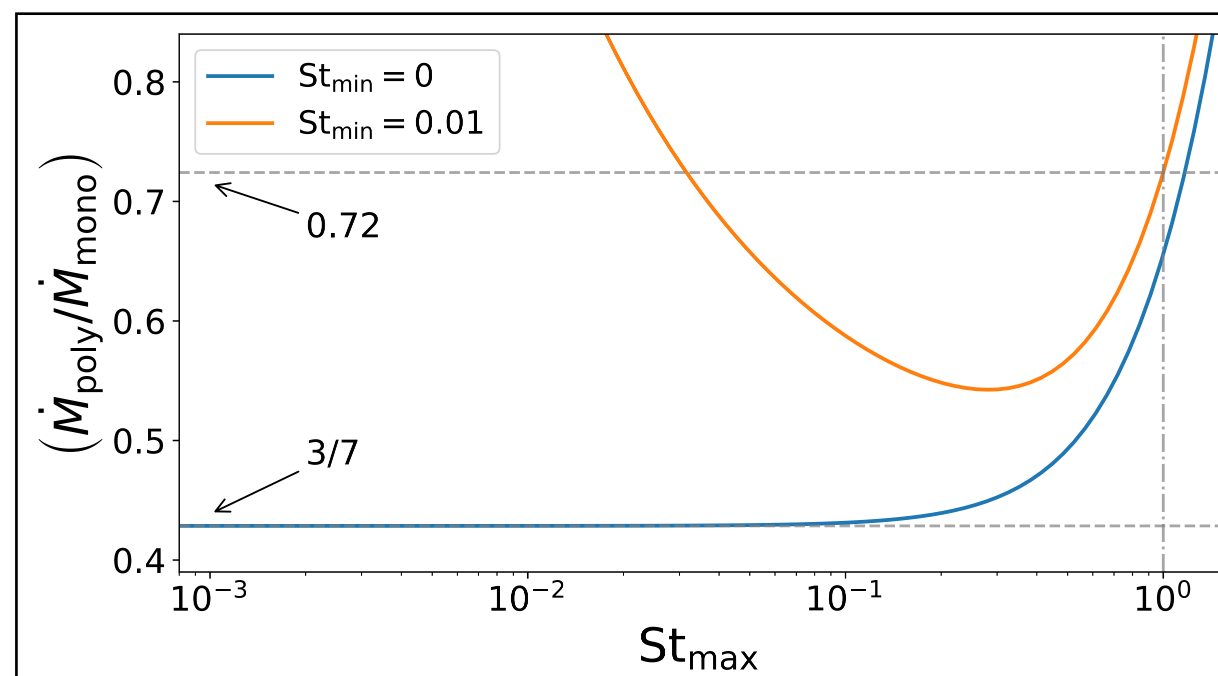
Received 2022 September 6; revised 2022 December 19; accepted 2022 December 29; published 2023 March 30

[Lyra et al. 2023]



[Konijn & Paardekooper 2026]

$$\left(\frac{\dot{M}_{\rm poly}}{\dot{M}_{\rm mono}}\right)_{\rm analytical} = \frac{3}{7}$$



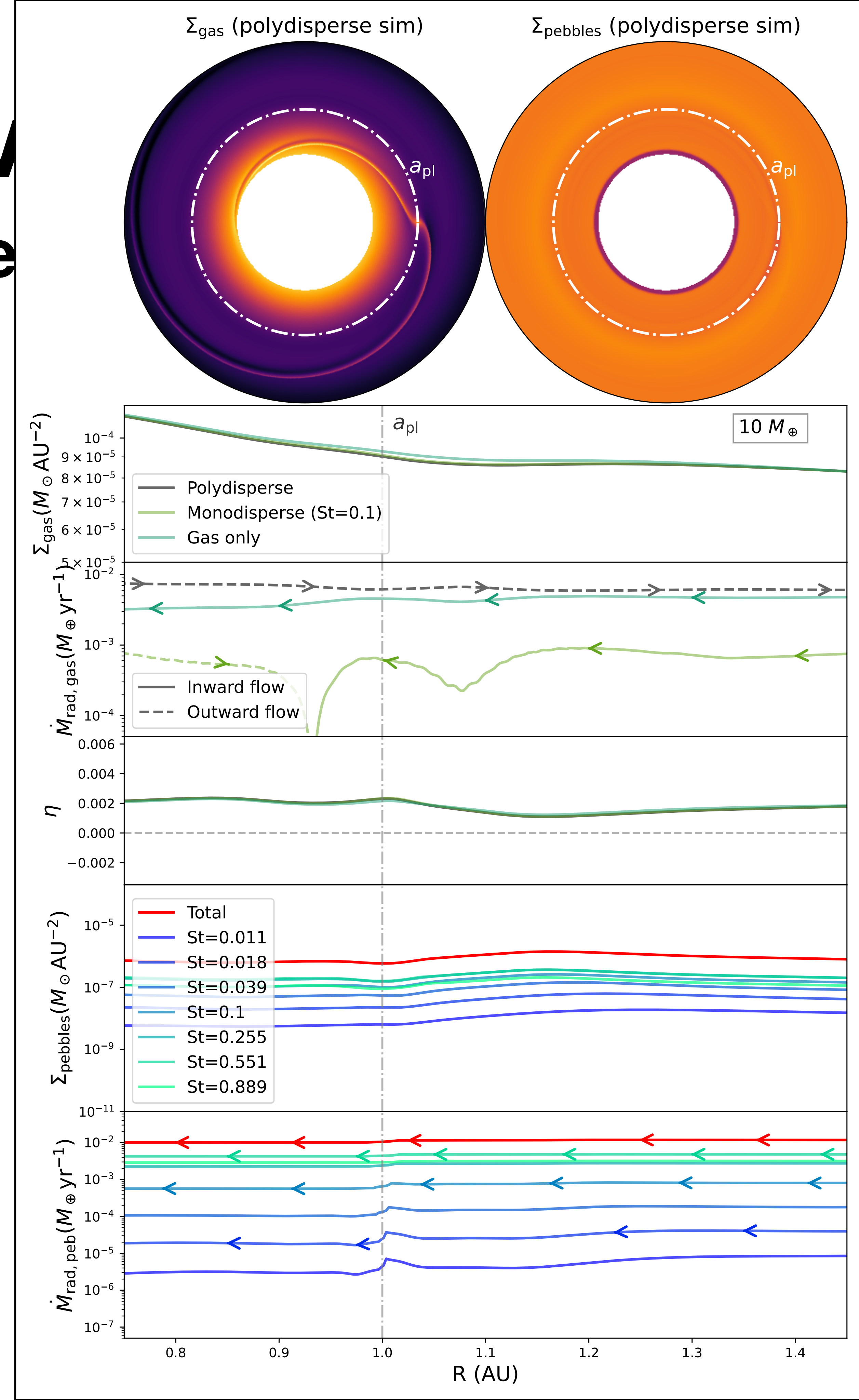
[Konijn & Paardekooper 2026]

Pebble accretion v

3. Watch the isolation proce

Lets just keep increasing mass

$\boxed{10\ M_\oplus}$

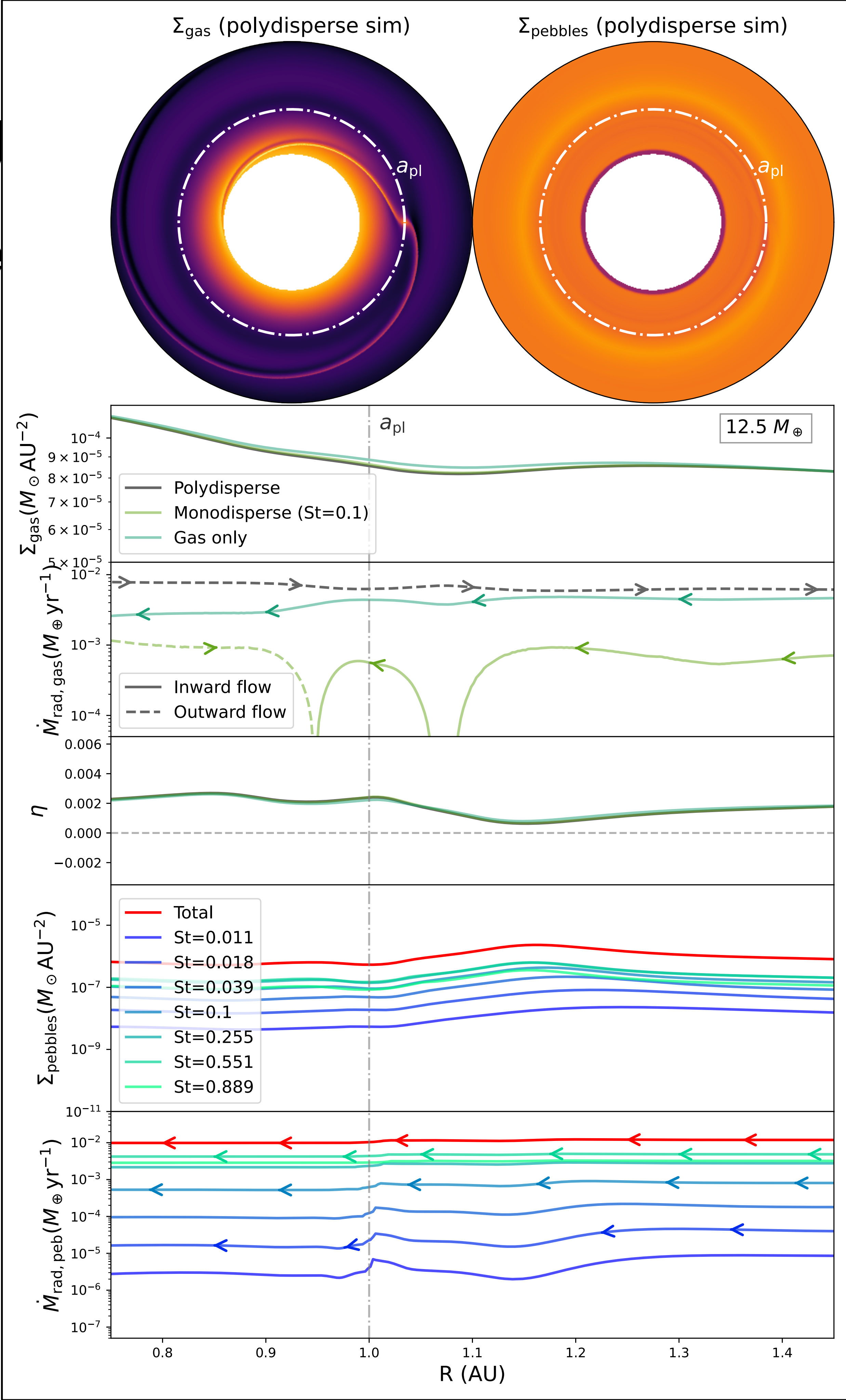


Pebble accretion v

3. Watch the isolation proce

Lets just keep increasing mass

$\boxed{12.5\ M_{\oplus}}$

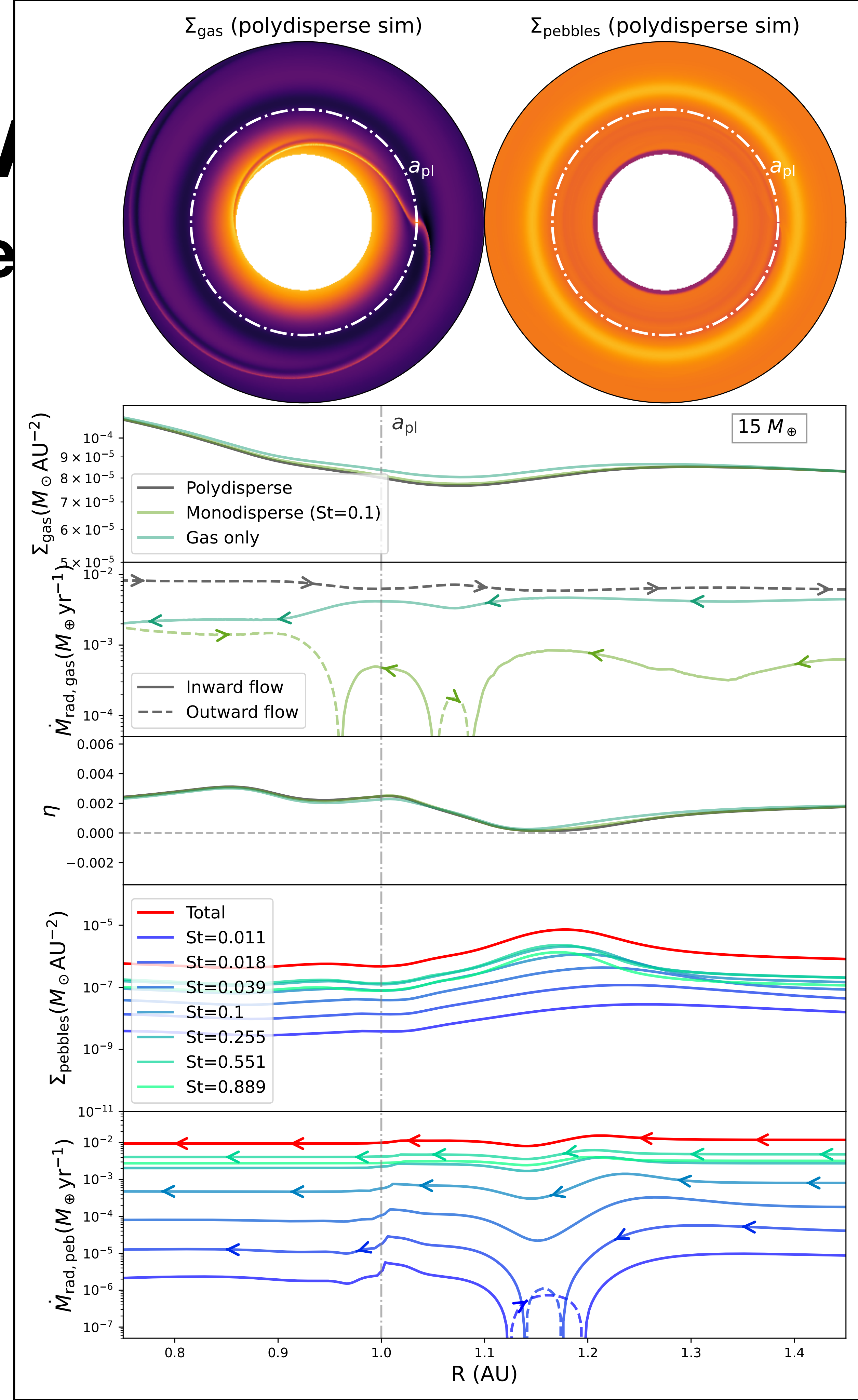


Pebble accretion v

3. Watch the isolation proce

Lets just keep increasing mass

$\boxed{15\ M_\oplus}$

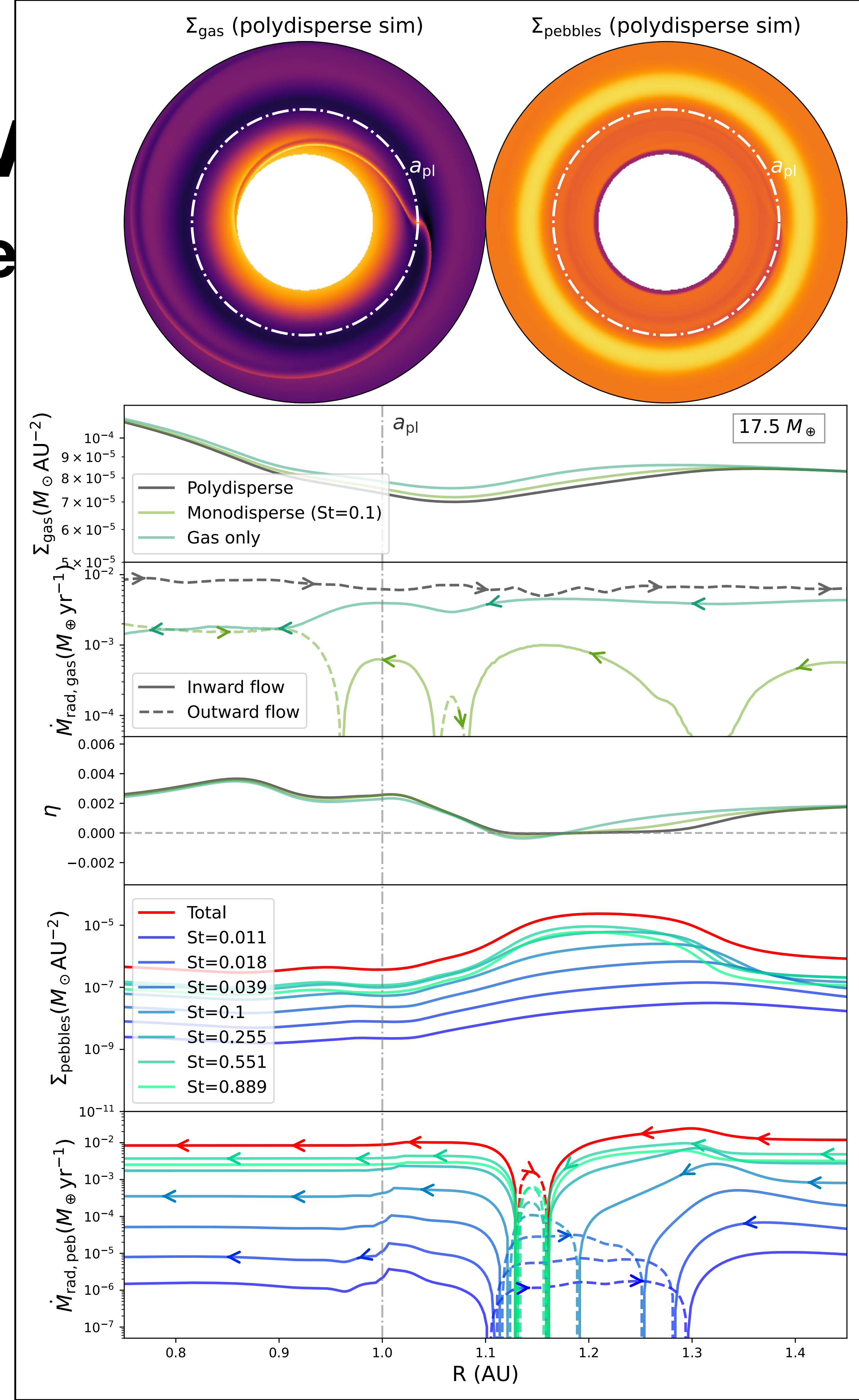


Pebble accretion v

3. Watch the isolation proce

Lets just keep increasing mass

$\boxed{17.5\ M_\oplus}$

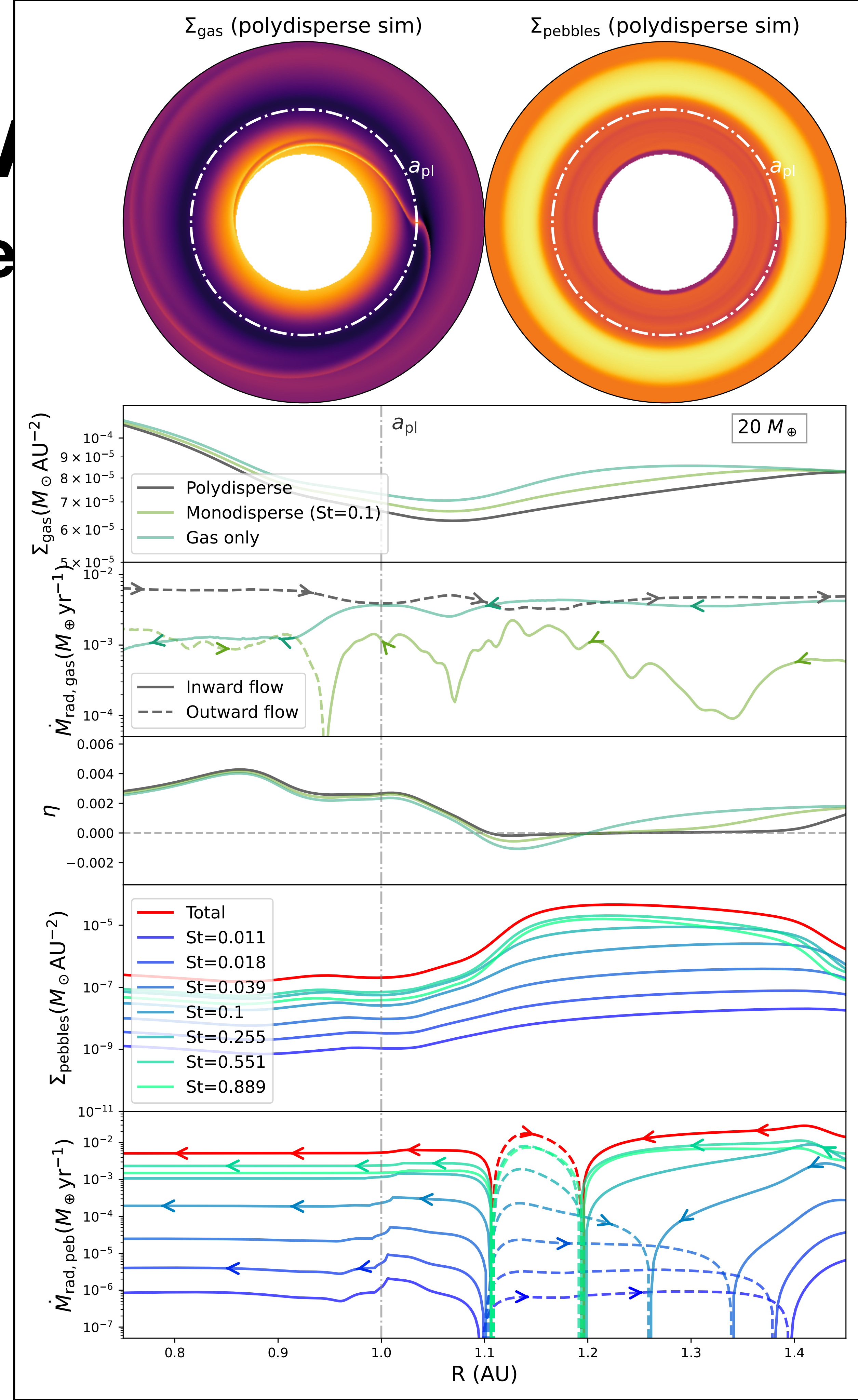


Pebble accretion v

3. Watch the isolation proce

Lets just keep increasing mass

$\boxed{20\ M_{\oplus}}$

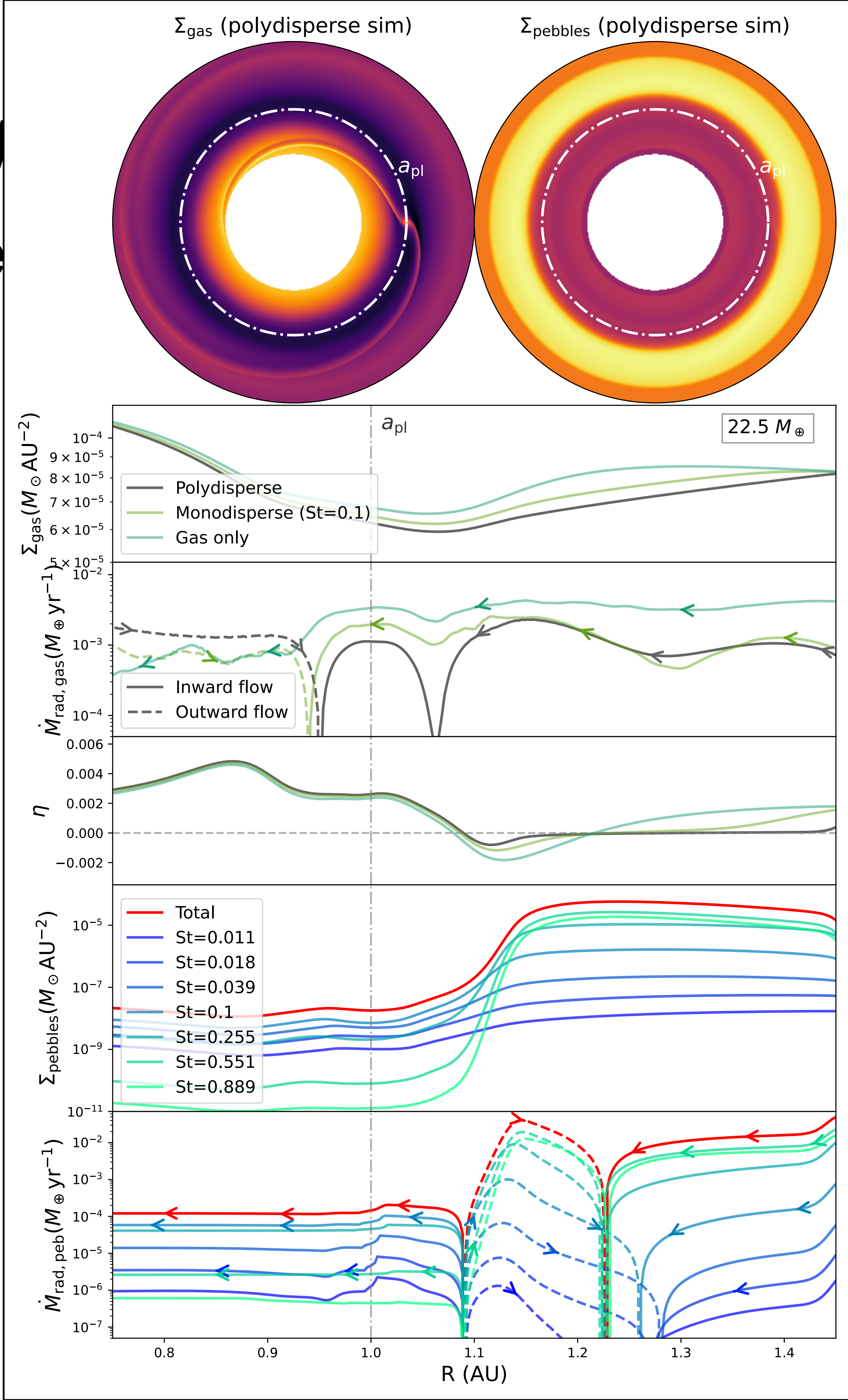


Pebble accretion v

3. Watch the isolation proce

Lets just keep increasing mass

$$\boxed{22.5\, M_\oplus}$$

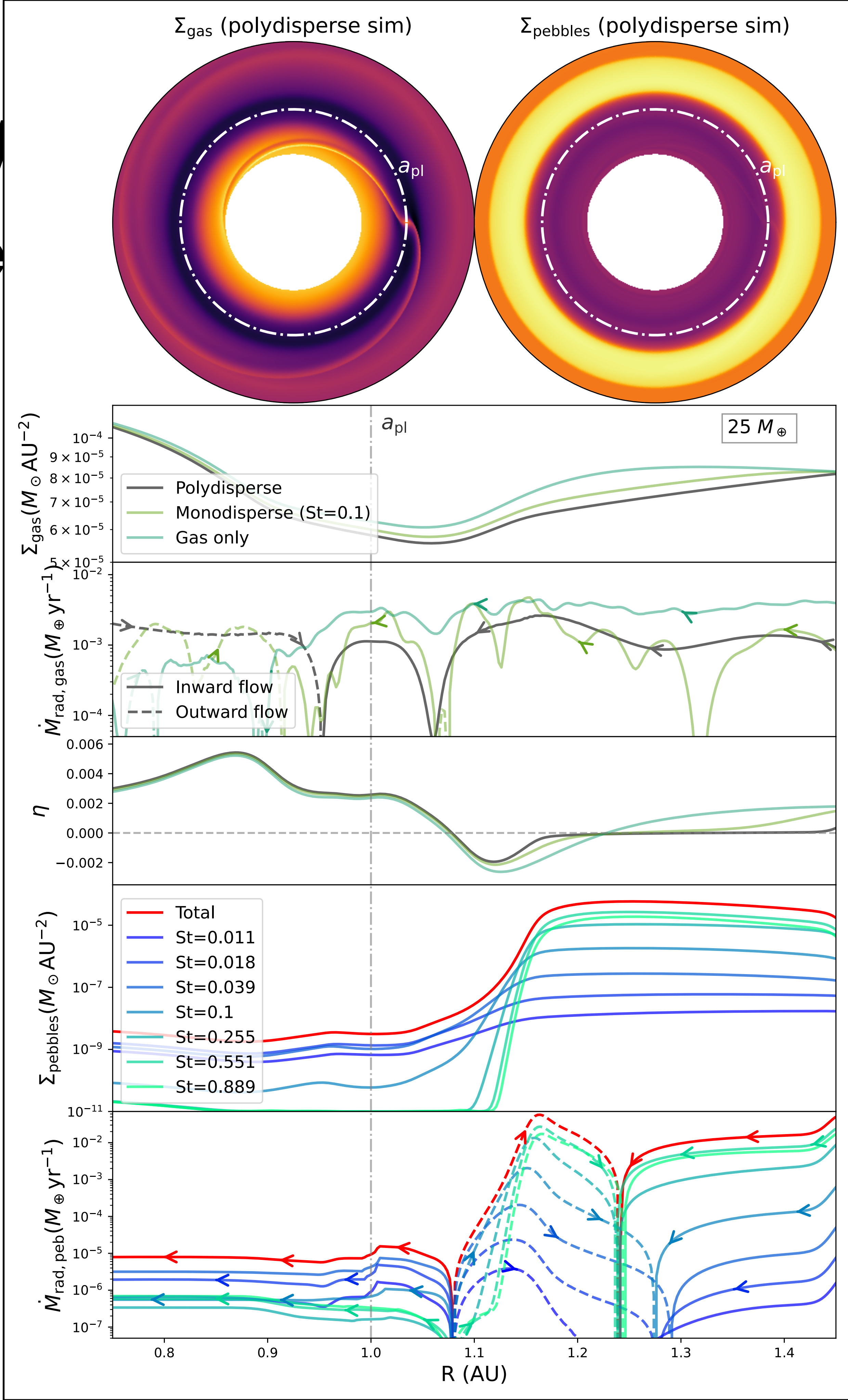


Pebble accretion v

3. Watch the isolation proce

Lets just keep increasing mass

$$\boxed{25\ M_\oplus}$$

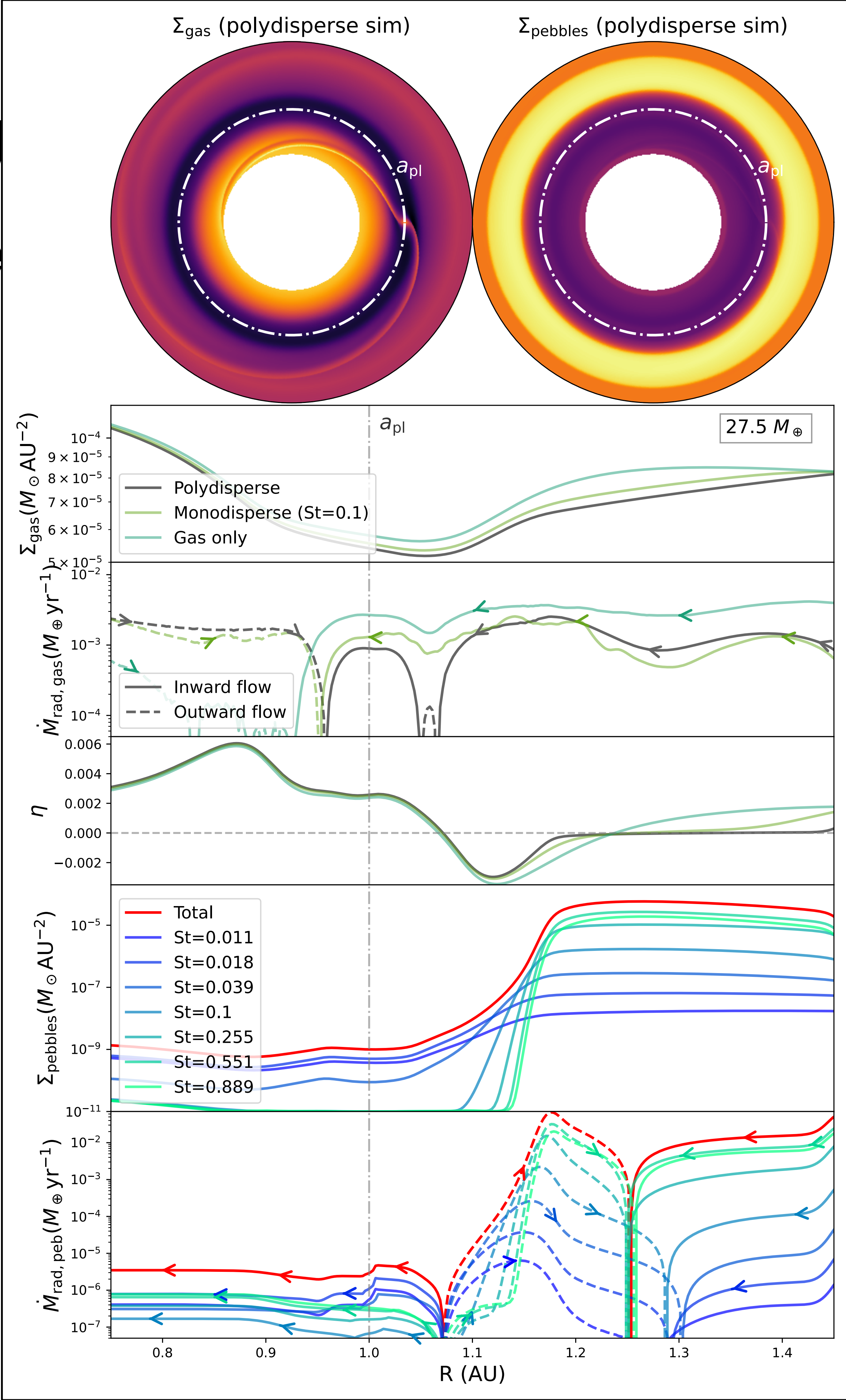


Pebble accretion v

3. Watch the isolation proce

Lets just keep increasing mass

$27.5\ M_\oplus$

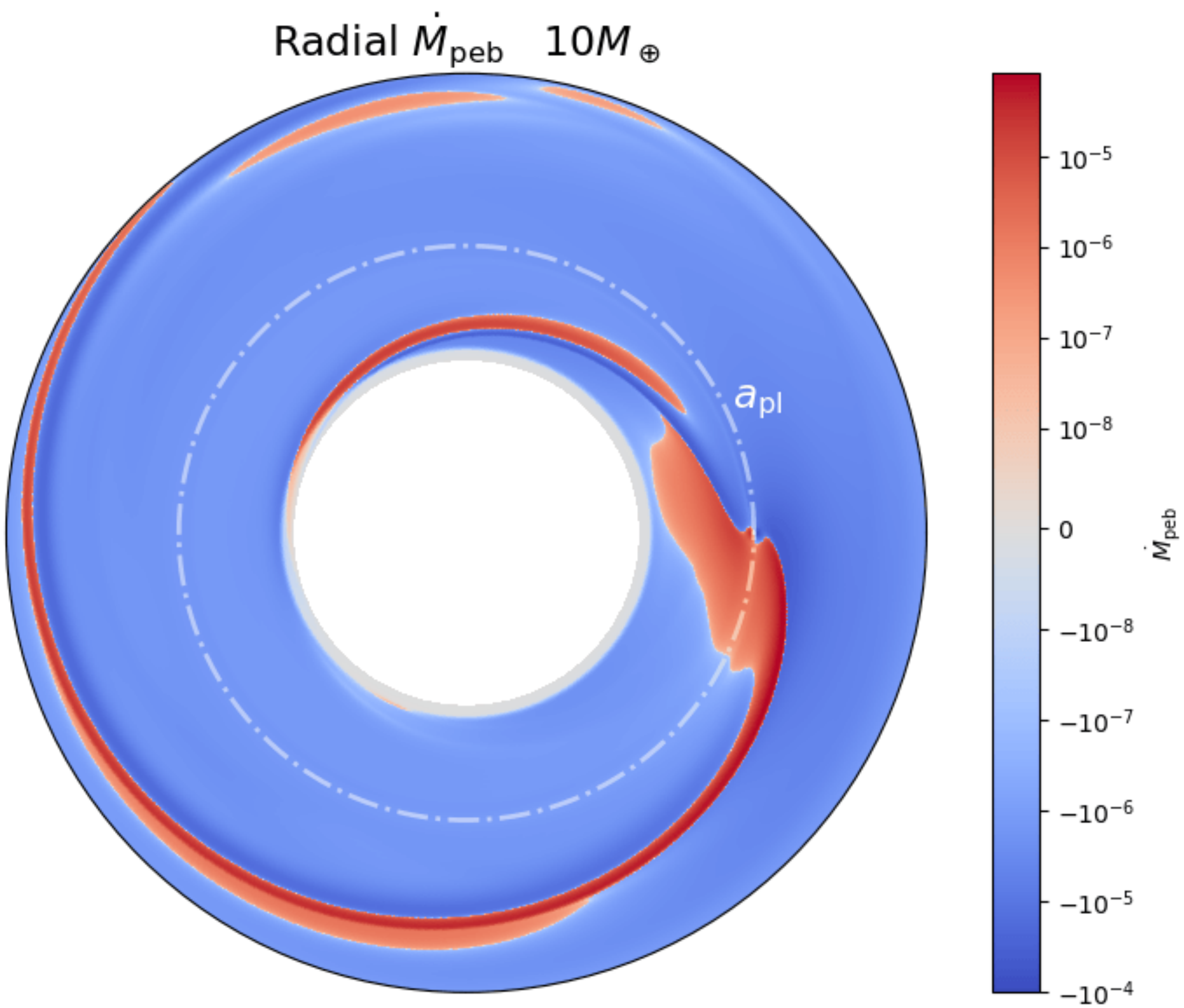


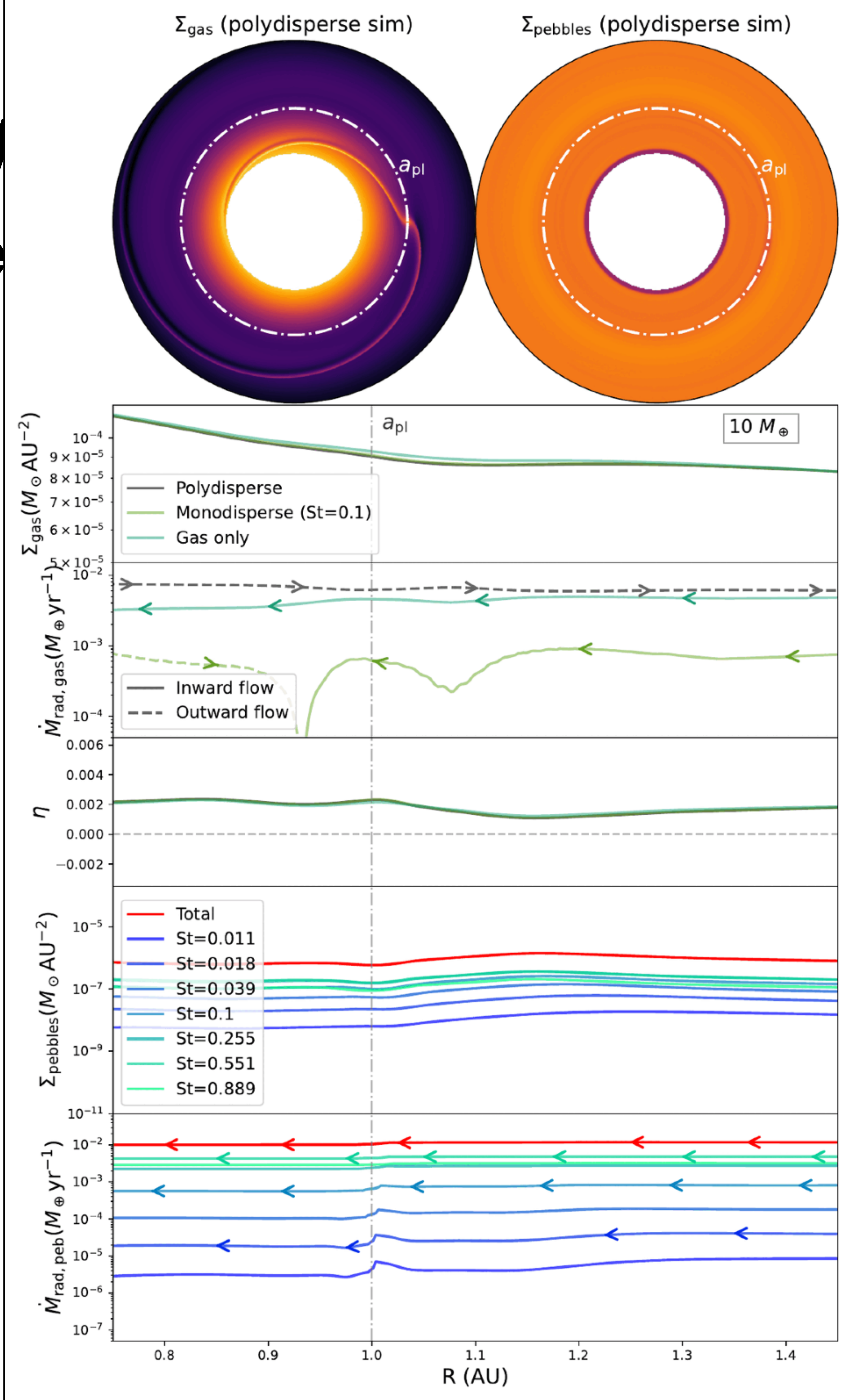
Pebble accretion v

3. Watch the isolation proce

Lets just keep increasing mass

- Radial dust flow: (movie)



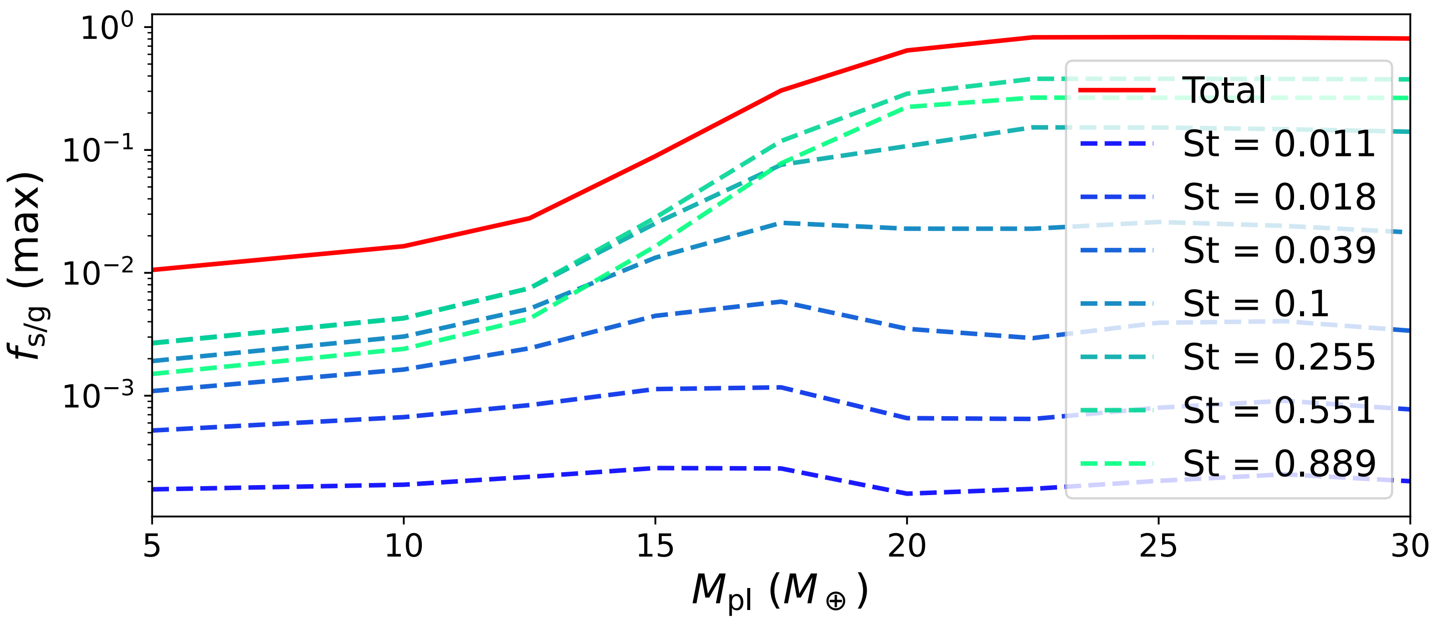


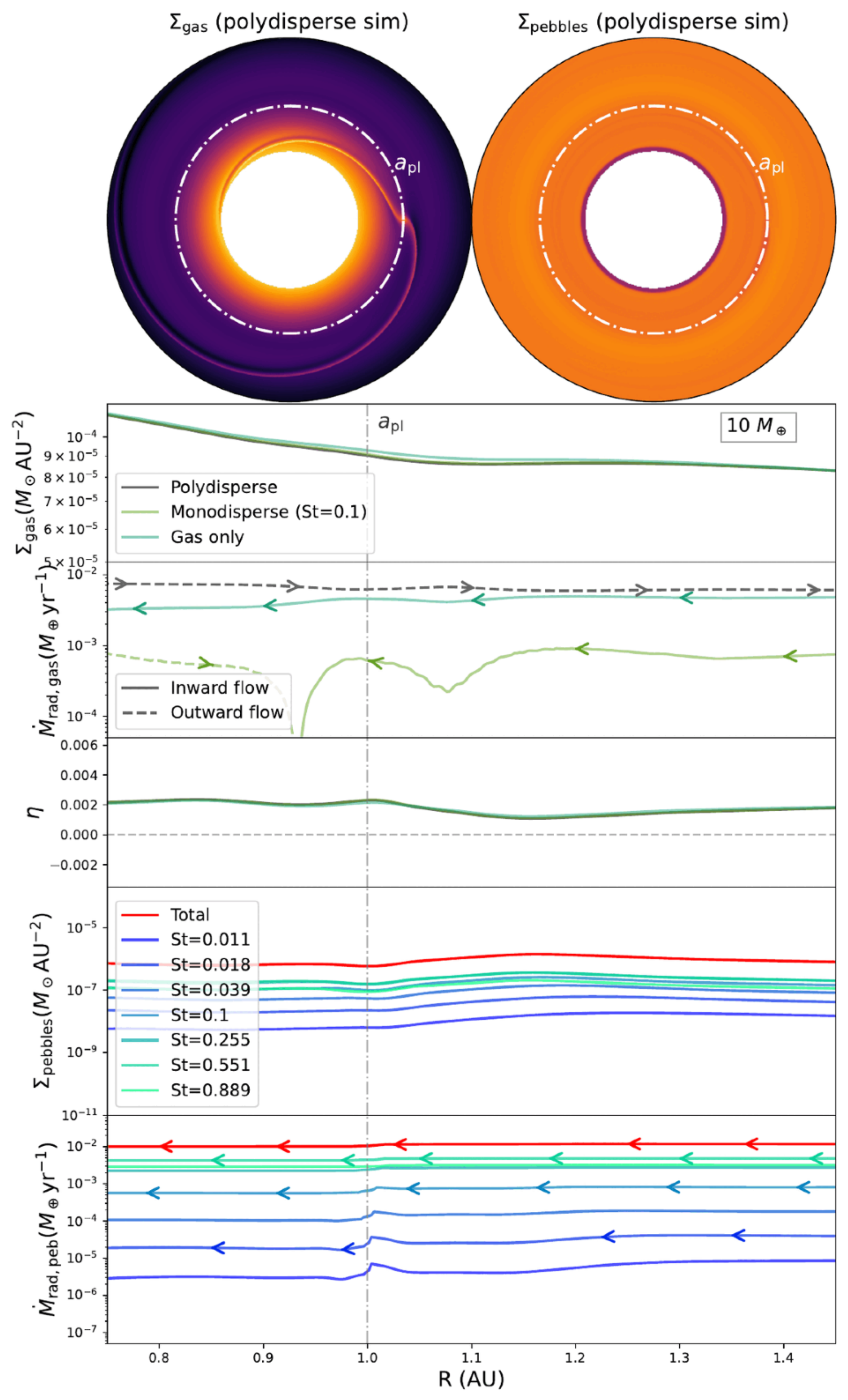
Pebble accretion v

3. Watch the isolation proce

Lets just keep increasing mass

- Dust-to-gas ratio increases to near unity values:





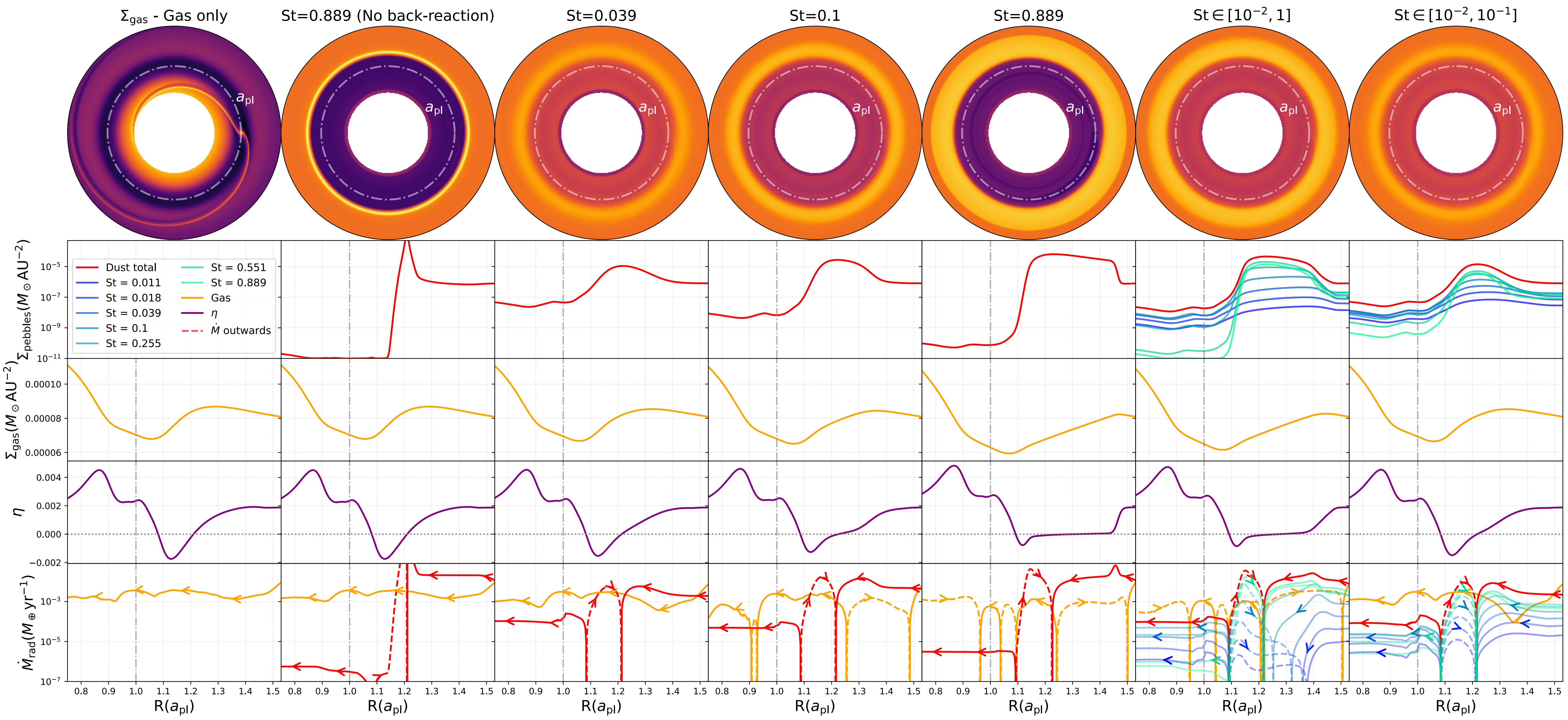
Pebble accretion via fluid method

3. Watch the isolation process

Lets just keep increasing mass

$$M_{\rm pl} = 22.5 M_\oplus$$

- Pebble distribution changes gas dynamics

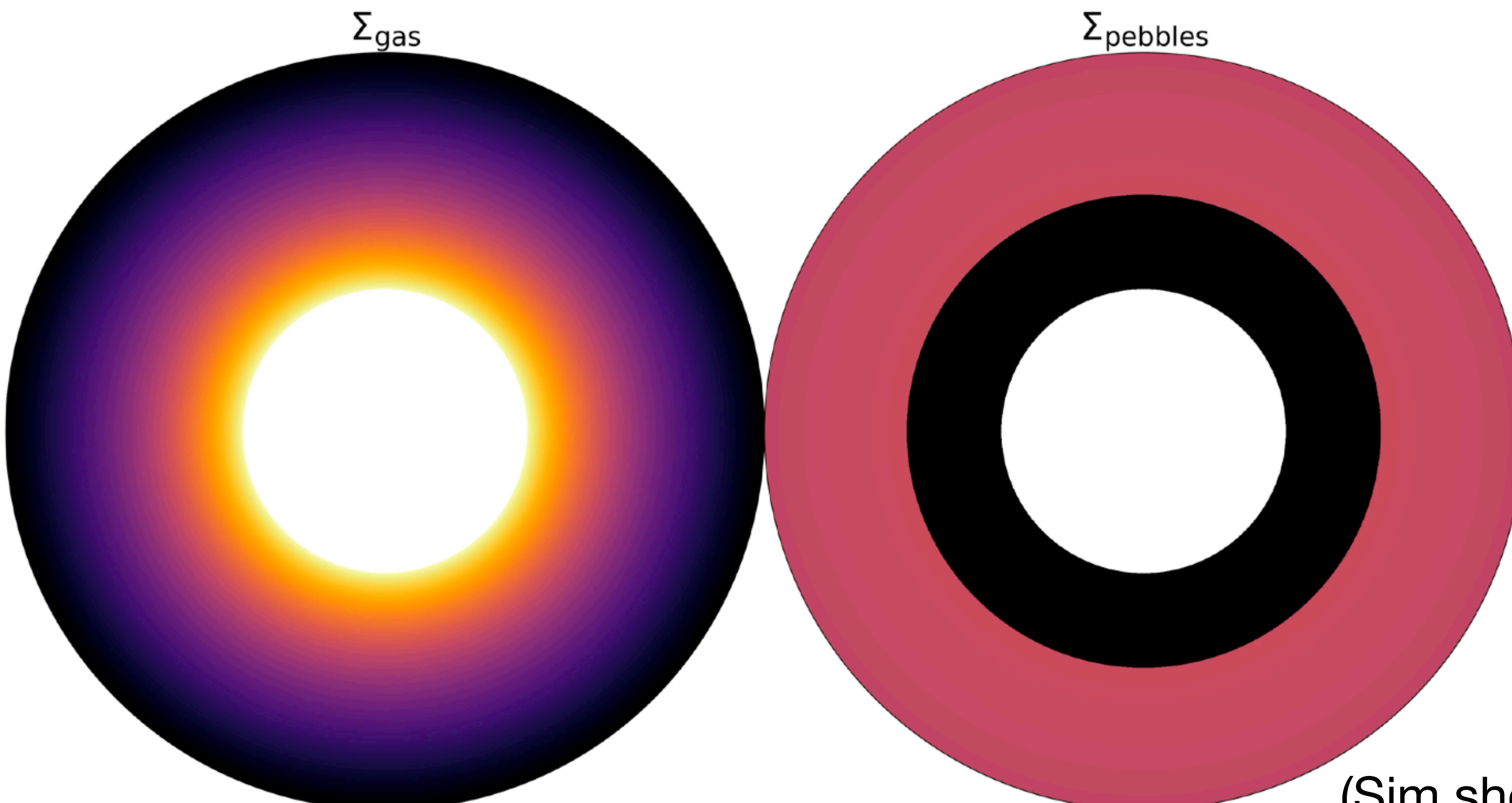


Pebble accretion via fluid method

3. Watch the isolation process

Lets just keep increasing mass

- Dust-to-gas ratio increases to near unity values:
- Which can also cause instabilities (movie)



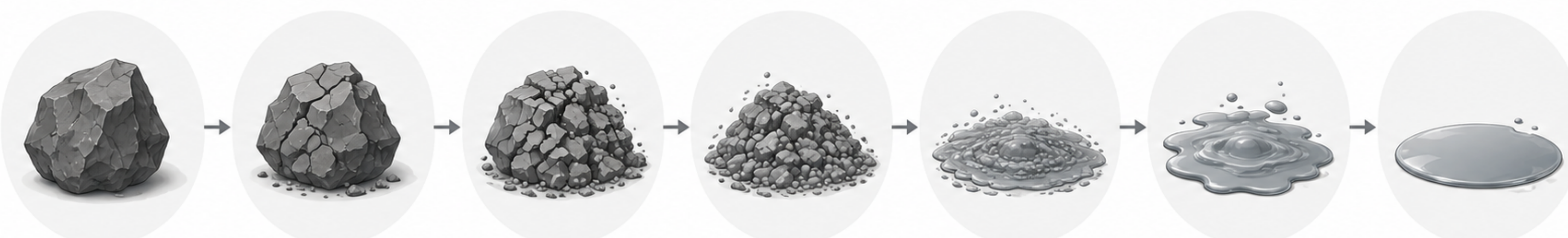
(Sim shows $\alpha \ll 10^{-3}$)

Thank you!

Implications of this talk

1. Pebble accretion is possible with a fluid approach
2. Pebble distribution changes gas dynamics drastically
3. After isolation, solid-to-gas ratios can reach unity values, which could cause instabilities
4. Potential for much more!
(isolation in 2.5D/3D, vertical flows, pebble torque, etc.)

(Part I now on arXiv!)



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